Om Namah shivay...

AMRITA VISHWA VIDYAPEETHAM | School of Engineering
Chennai Campus

23MEE203_Metallurgy and Materials Science

Phase diagrams:

Phase diagrams provide a graphical means of presenting the results of experimental studies of complex natural processes, such that at a given temperature and pressure for a specific system at equilibrium the phase or phases present can be determined.

EQUILIBRIUM - The condition of minimum energy for the system such that the state of a reaction will not change with time provided that pressure and temperature are kept constant.

Component - components are pure metals and/or compounds of which an alloy is composed.

- Components may be oxides, elements or minerals, depending on the system being examined.

For example, experiments carried out in the H_2O system, show that the phases which appear over a wide temperature and pressure range are ice, liquid water and water vapour. The composition of each phase is H_2O and only one chemical parameter or component is required to describe the composition of each phase.

Another example is copper-zinc brass, the components are Cu and Zn.

System

The system may refer to a specific body of material under consideration (e.g., a ladle of molten steel), or it may relate to the series of possible alloys consisting of the same components but without regard to alloy composition (e.g., the iron–carbon system).

Solute and solvent

A solid solution consists of atoms of at least two different types; the solute atoms occupy either substitutional or interstitial positions in the solvent lattice, and the crystal structure of the solvent is maintained.

Phase

A phase may be defined as a homogeneous portion of a system that has uniform physical and chemical characteristics. Every pure material is considered to be a phase; so also is every solid, liquid, and gaseous solution. For example, the sugar–water syrup solution just discussed is one phase, and solid sugar is another.

For many alloy systems and at some specific temperature, there is a maximum concentration of solute atoms that may dissolve in the solvent to form a solid solution; this is called a *solubility limit*.

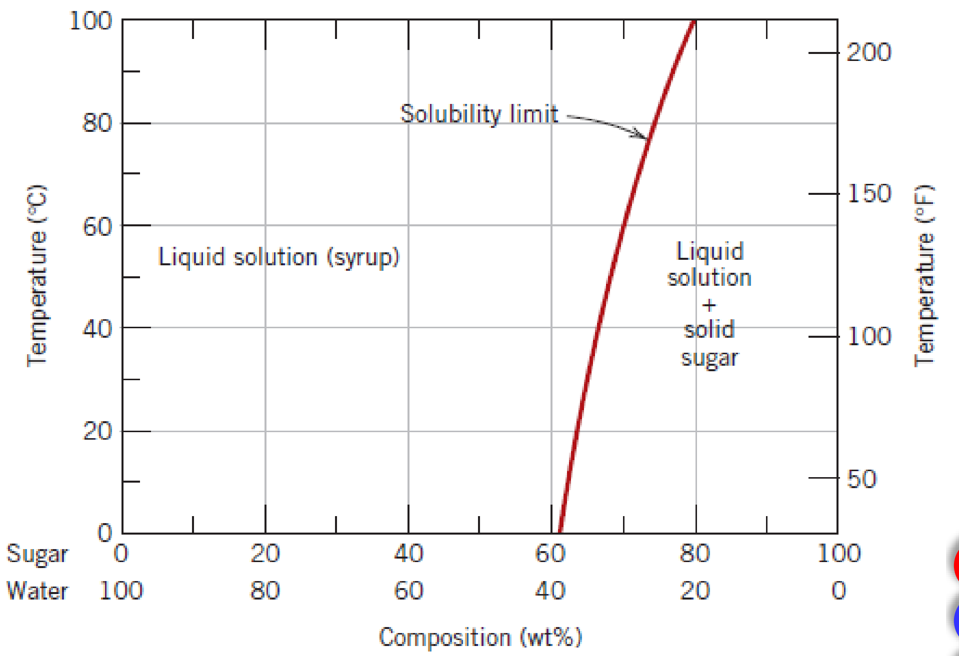


Figure 1 The solubility of sugar (C12H22O11) in a sugar–water syrup.

PHASE EQUILIBRIA

Equilibrium is another essential concept; it is best described in terms of a thermodynamic quantity called **free energy.** In brief, *free energy* is a function of the internal energy of a system and also the randomness or disorder of the atoms or molecules (or *entropy*). A system is at equilibrium if its free energy is at a minimum under some specified combination of temperature, pressure, and composition.

Phase equilibrium

The term **phase equilibrium,** often used in the context of this discussion, refers to equilibrium as it applies to systems in which more than one phase may exist. Phase equilibrium is reflected by a constancy with time in the phase characteristics of a system.

Perhaps an example best illustrates this concept. Suppose that a sugar–water syrup is contained in a closed vessel and the solution is in contact with solid sugar at 20°C. If the system is at equilibrium, the composition of the syrup is 65 wt% $C_{12}H_{22}O_{11}$–35 wt% H_2O (Figure 1), and the amounts and compositions of the syrup and solid sugar will remain constant with time. If the temperature of the system is suddenly raised—say, to 100°C—this equilibrium or balance is temporarily upset and the solubility limit is increased to 80 wt% $C_{12}H_{22}O_{11}$ (Figure 1). Thus, some of the solid sugar will go into solution in the syrup. This will continue until the new equilibrium syrup concentration is established at the higher temperature.

- Free energy considerations and diagrams similar to Figure 1 provide information about the equilibrium characteristics of a particular system, which is important, but they do not indicate the time period necessary for the attainment of a new equilibrium state.

- It is often the case, especially in solid systems, that a state of equilibrium is never completely achieved because the rate of approach to equilibrium is extremely slow; such a system is said to be in a nonequilibrium or metastable state.

- A metastable state or microstructure may persist indefinitely, experiencing only extremely slight and almost imperceptible changes as time progresses.

- Often, metastable structures are of more practical significance than equilibrium ones.

- For example, some steel and aluminum alloys rely on their strength in the development of metastable microstructures during carefully designed heat treatments.

- Thus, it is important to understand not only equilibrium states and structures but also the speed or rate at which they are established and the factors that affect the rate.

ONE-COMPONENT (OR UNARY) PHASE DIAGRAMS

Much of the information about the control of the phase structure of a particular system is conveniently and concisely displayed in what is called a **phase diagram,** also often termed an *equilibrium diagram.* Three externally controllable parameters affect phase structure—temperature, pressure, and composition—and phase diagrams are constructed when various combinations of these parameters are plotted against one another.

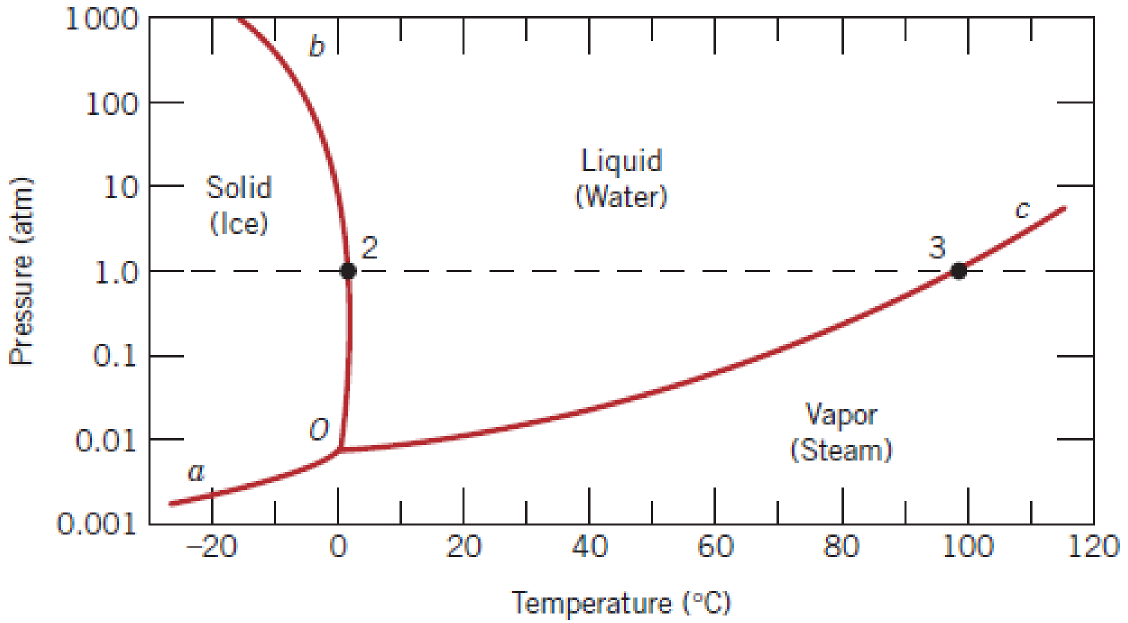


Figure 2 Pressure–temperature phase diagram for H_2O. The intersection of the dashed horizontal line at 1 atm pressure with the solid-liquid phase boundary (point 2) corresponds to the melting point at this pressure (T = 0°C). Similarly, point 3, the intersection with the liquid–vapor boundary, represents the boiling point (T = 100°C). simple cubic unit cell

Binary Phase Diagrams

BINARY ISOMORPHOUS SYSTEMS

The composition ranges from 0 wt% Ni (100 wt% Cu) on the far left horizontal extreme to 100 wt% Ni (0 wt% Cu) on the right. Three different phase regions, or *fields*, appear on the diagram—an alpha (α) field, a liquid (L) field, and a two-phase $\alpha + L$ field. Each region is defined by the phase or phases that exist over the range of temperatures and compositions delineated by the phase boundary lines.

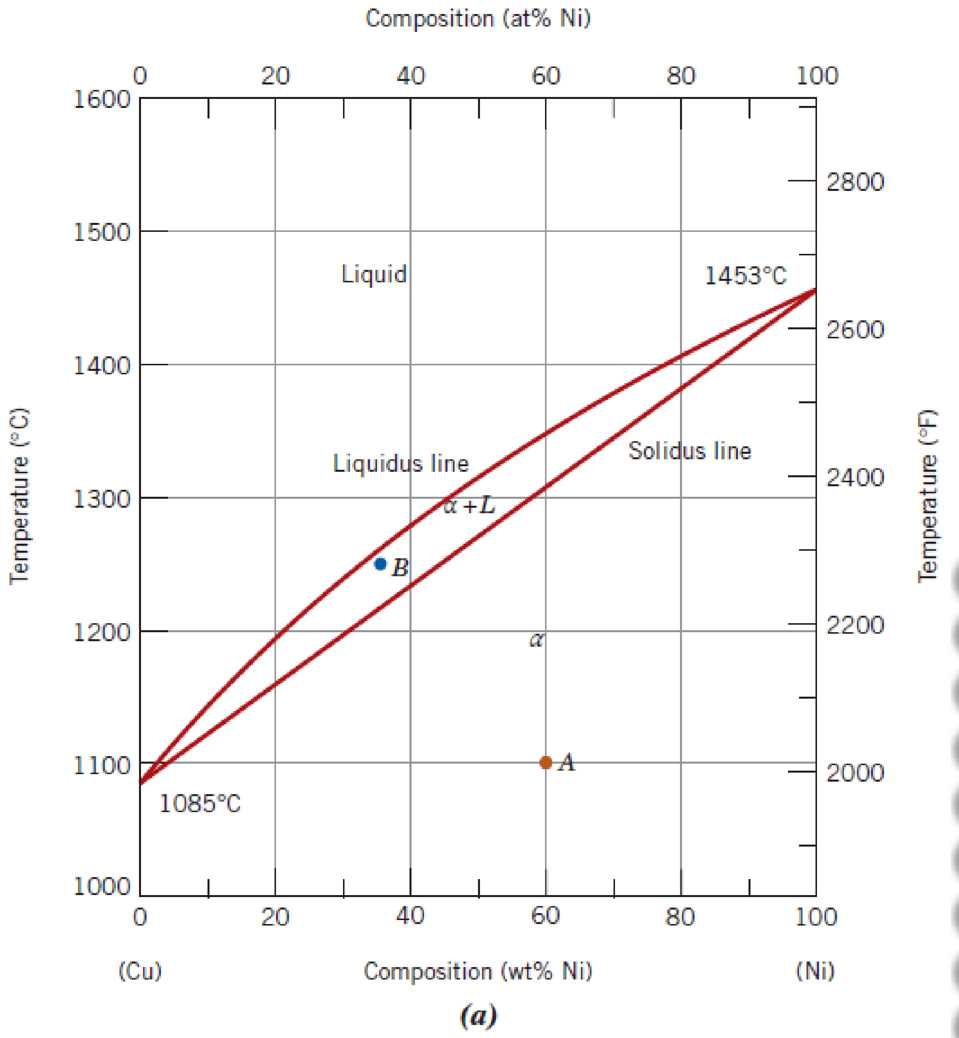


Figure 3 (a) The copper–nickel phase diagram.

Phases Present

The establishment of what phases are present is relatively simple.

For example, an alloy of composition 60 wt% Ni–40 wt% Cu at 1100°C would be located at point *A* in Figure 3a; because this is within the *α* region, only the single *α* phase will be present. However, a 35 wt% Ni–65 wt% Cu alloy at 1250°C (point *B*) consists of both *α* and liquid phases at equilibrium.

Determination of Phase Compositions

For example, consider the 60 wt% Ni–40 wt% Cu alloy at 1100°C (point *A*, Figure 3a). At this composition and temperature, only the *α* phase is present, having a composition of 60 wt% Ni–40 wt% Cu.

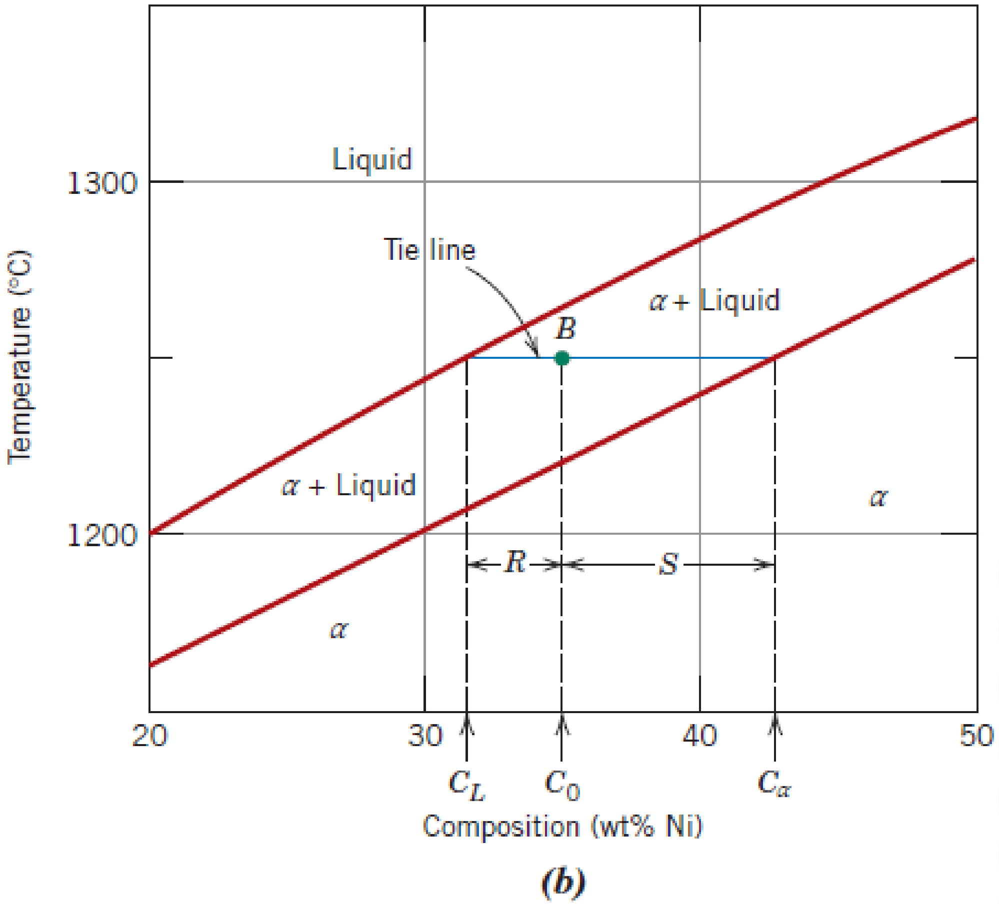


Figure 3 (b) A portion of the copper-nickel phase diagram for which compositions and phase amounts are determined at point B.

To compute the equilibrium concentrations of the two phases, the following procedure is used:

- A tie line is constructed across the two-phase region at the temperature of the alloy.
- The intersections of the tie line and the phase boundaries on either side are noted.
- Perpendiculars are dropped from these intersections to the horizontal composition axis, from which the composition of each of the respective phases is read.

For example, consider again the 35 wt% Ni–65 wt% Cu alloy at 1250°C, located at point *B* in Figure 3b and lying within the $\alpha + L$ region.

Thus, the problem is to determine the composition (in wt% Ni and Cu) for both the α and liquid phases.

The tie line is constructed across the $\alpha + L$ phase region, as shown in Figure 3b.

The perpendicular from the intersection of the tie line with the liquidus boundary meets the composition axis at 31.5 wt% Ni–68.5 wt% Cu, which is the composition of the liquid phase, C_L.

Likewise, for the solidus–tie line intersection, we find a composition for the α solid-solution phase, C_α, of 42.5 wt% Ni–57.5 wt% Cu.

Determination of Phase Amounts

From the previous example for the 60 wt% Ni–40 wt% Cu alloy at 1100°C (point *A* in Figure 3a), only the α phase is present; hence, the alloy is completely, or 100%, α. If the composition and temperature position is located within a two-phase region, things are more complex. The tie line must be used in conjunction with a procedure that is often called the **lever rule** (or the *inverse lever rule*), which is applied as follows:

1. The tie line is constructed across the two-phase region at the temperature of the alloy.
2. The overall alloy composition is located on the tie line.
3. The fraction of one phase is computed by taking the length of the tie line from the overall alloy composition to the phase boundary for the *other* phase and dividing by the total tie-line length.
4. The fraction of the other phase is determined in the same manner.
5. If phase percentages are desired, each phase fraction is multiplied by 100. When the composition axis is scaled in weight percent, the phase fractions computed using the lever rule are *mass fractions*—the mass (or weight) of a specific phase divided by the total alloy mass (or weight). The mass of each phase is computed from the product of each phase fraction and the total alloy mass.

Consider again the example shown in Figure 3*b*, in which at 1250°C both α and liquid phases are present for a 35 wt% Ni–65 wt% Cu alloy. The problem is to compute the fraction of each of the α and liquid phases. The tie line is constructed that was used for the determination of α and L phase compositions. Let the overall alloy composition be located along the tie line and denoted as C_0, and let the mass fractions be represented by W_L and W_α for the respective phases. From the lever rule, W_L may be computed according to

$$W_L = \frac{S}{R + S}$$

or, by subtracting compositions,

$$W_L = \frac{C_\alpha - C_0}{C_\alpha - C_L}$$

The composition need be specified in terms of only one of the constituents for a binary alloy; for the preceding computation, weight percent nickel is used (i.e., C_0 = 35 wt% Ni, C_α = 42.5 wt% Ni, and C_L = 31.5 wt% Ni), and

$$W_L = \frac{42.5 - 35}{42.5 - 31.5} = 0.68$$

Similarly, for the α phase,

$$W_\alpha = \frac{R}{R + S} = \frac{C_0 - C_L}{C_\alpha - C_L}$$

$$W_\alpha = \frac{35 - 31.5}{42.5 - 31.5} = 0.32$$

Practice example:

1. A copper–nickel alloy of composition 70 wt% Ni–30 wt% Cu is slowly heated from a temperature of 1300°C (2370°F).
   a) At what temperature does the first liquid phase form?
   b) What is the composition of this liquid phase?
   c) At what temperature does complete melting of the alloy occur?
   d) What is the composition of the last solid remaining prior to complete melting?

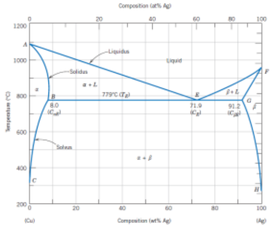
2. A 50 wt% Ni–50 wt% Cu alloy is slowly cooled from 1400°C (2550°F) to 1200°C (2190°F).
   a) At what temperature does the first solid phase form?
   b) What is the composition of this solid phase?
   c) At what temperature does the liquid solidify?
   d) What is the composition of this last remaining liquid phase?

Practice example:

3. A 65 wt% Ni–35 wt% Cu alloy is heated to a temperature within the α + liquid-phase region. If the composition of the α phase is 70 wt% Ni, determine:
    a) The temperature of the alloy
    b) The composition of the liquid phase
    c) The mass fractions of both phases

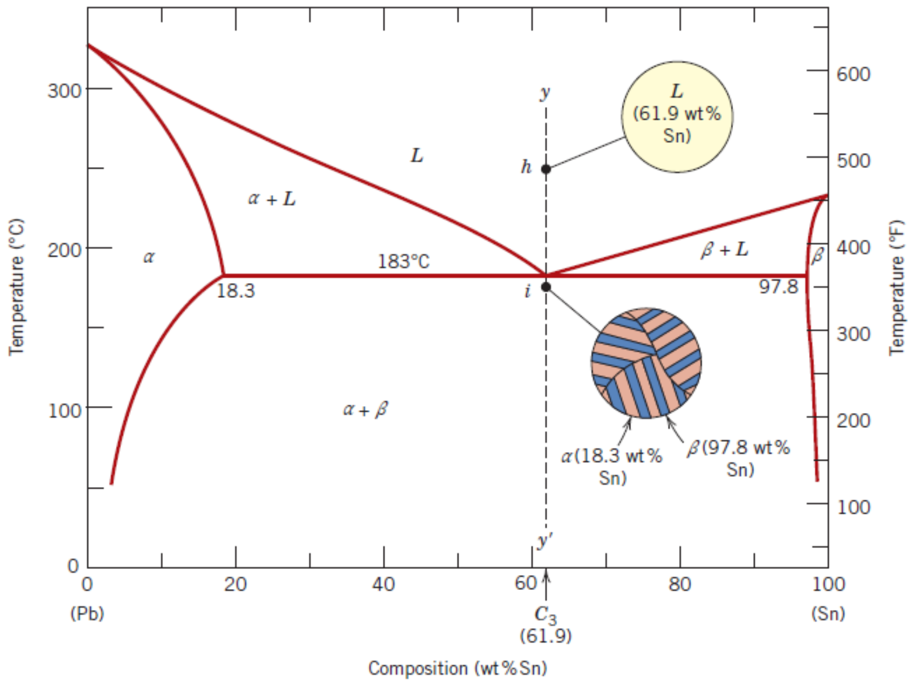
A 90 wt% Ag- 10 wt% Cu alloy is heated to a temperature within the b + liquid phase region. If the composition of the liquid phase is 85 wt% Ag, determine:

(a) The temperature of the alloy
(b) The composition of the b phase
(c) The mass fractions of both phases



Practice example:

4. A lead–tin alloy of composition 30 wt% Sn–70 wt% Pb is slowly heated from a temperature of 150°C (300°F).
   a) At what temperature does the first liquid phase form?
   b) What is the composition of this liquid phase?
   c) At what temperature does complete melting of the alloy occur?
   d) What is the composition of the last solid remaining prior to complete melting?



DEVELOPMENT OF MICROSTRUCTURE IN ISOMORPHOUS ALLOYS

Equilibrium Cooling

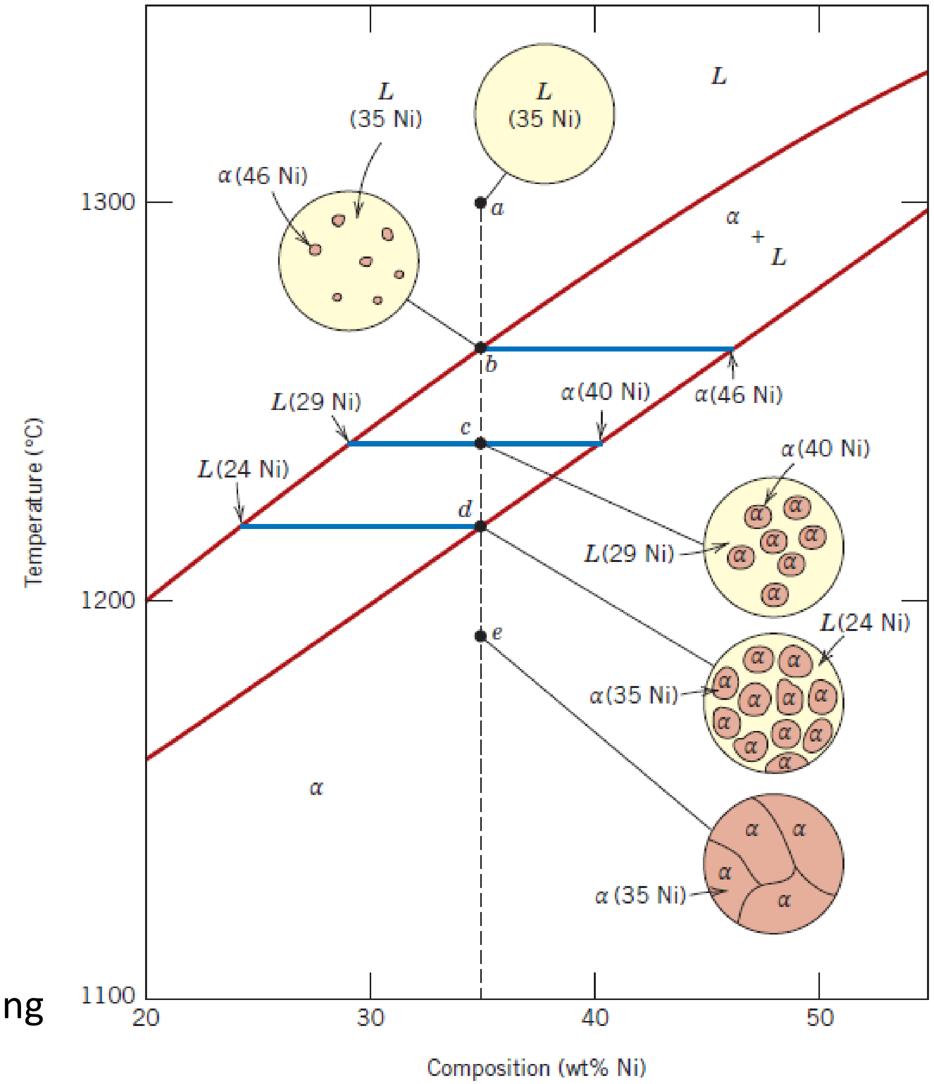


Figure 4 Schematic representation of the development of microstructure during the equilibrium solidification of a 35 wt% Ni–65 wt% Cu alloy.

BINARY EUTECTIC SYSTEMS

First, three single-phase regions are found on the diagram: *α*, *β*, and liquid. The *α* phase is a solid solution rich in copper; it has silver as the solute component and an FCC crystal structure. The *β*-phase solid solution also has an FCC structure, but copper is the solute. Pure copper and pure silver are also considered to be *α* and *β* phases, respectively.

Thus, the solubility in each of these solid phases is limited, in that at any temperature below line *BEG* only a limited concentration of silver dissolves in copper (for the *α* phase), and similarly for copper in silver (for the *β* phase).

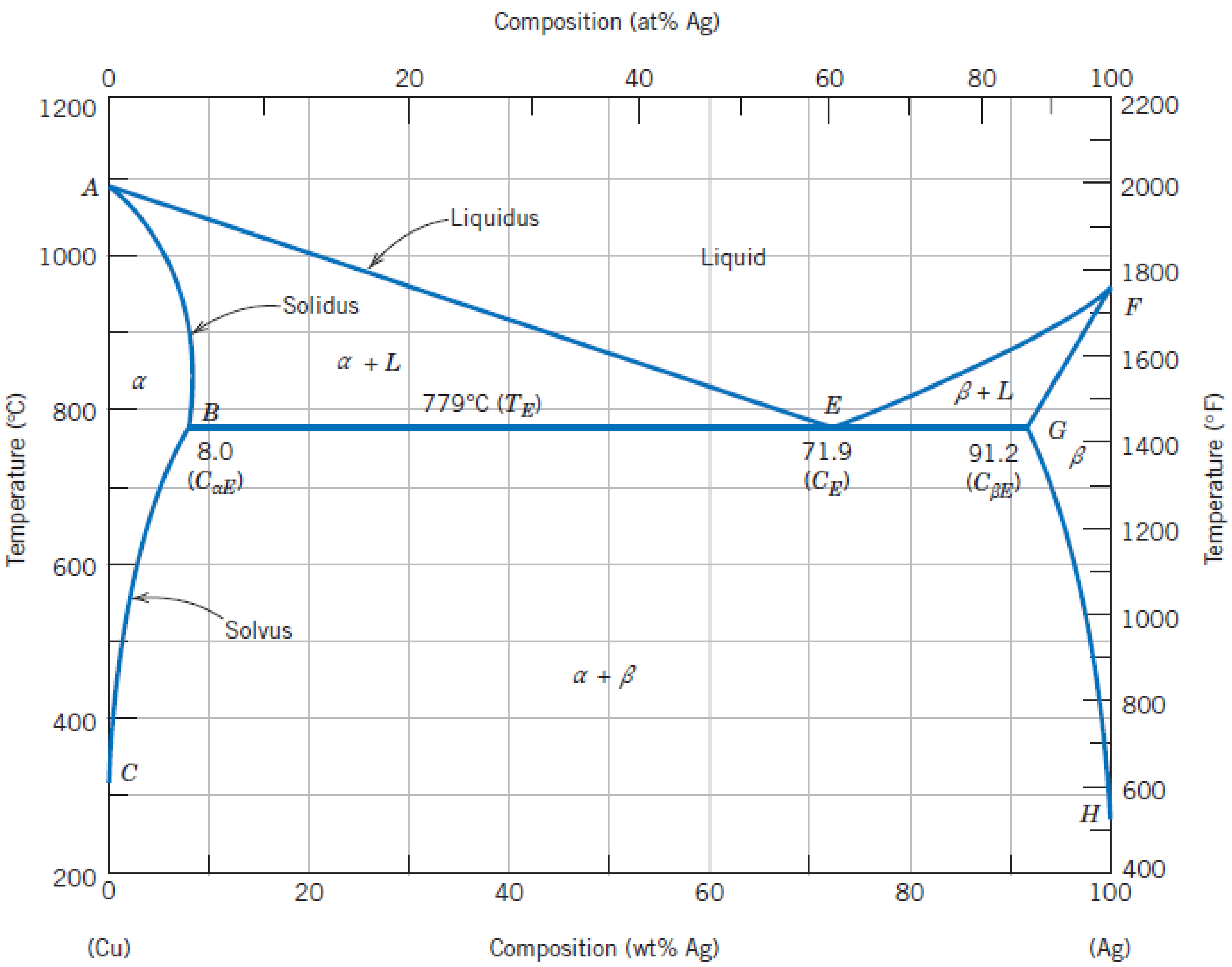


Figure 5 The copper–silver phase diagram.

An important reaction occurs for an alloy of composition C_E as it changes the temperature in passing through T_E; this reaction may be written as follows:

$$L(C_E) \underset{\text{heating}}{\overset{\text{cooling}}{\rightleftharpoons}} \alpha(C_{\alpha E}) + \beta(C_{\beta E})$$

Or, upon cooling, a liquid phase is transformed into the two solid α and β phases at the temperature T_E; the opposite reaction occurs upon heating. This is called a **eutectic reaction** (*eutectic* means "easily melted"), and point *E* on the diagram is called the *eutectic point;* furthermore, C_E and T_E represent the eutectic composition and temperature, respectively. As also noted in Figure 5, $C_{\alpha E}$ and $C_{\beta E}$ are the respective compositions of the α and β phases at T_E. Thus, for the copper-silver system, the eutectic reaction may also be written as follows:

$$L(71.9 \text{ wt\% Ag}) \underset{\text{heating}}{\overset{\text{cooling}}{\rightleftharpoons}} \alpha(8.0 \text{ wt\% Ag}) + \beta(91.2 \text{ wt\% Ag})$$

This eutectic reaction is termed an *invariant reaction* inasmuch as it occurs under equilibrium conditions at a specific temperature (T_E) and specific compositions (C_E, $C_{\alpha E}$ and $C_{\beta E}$), which are constant (i.e., invariable) for a specific binary system.1 Furthermore, the horizontal solidus line *BEG* at T_E is sometimes called the *eutectic isotherm*.

Practice example:

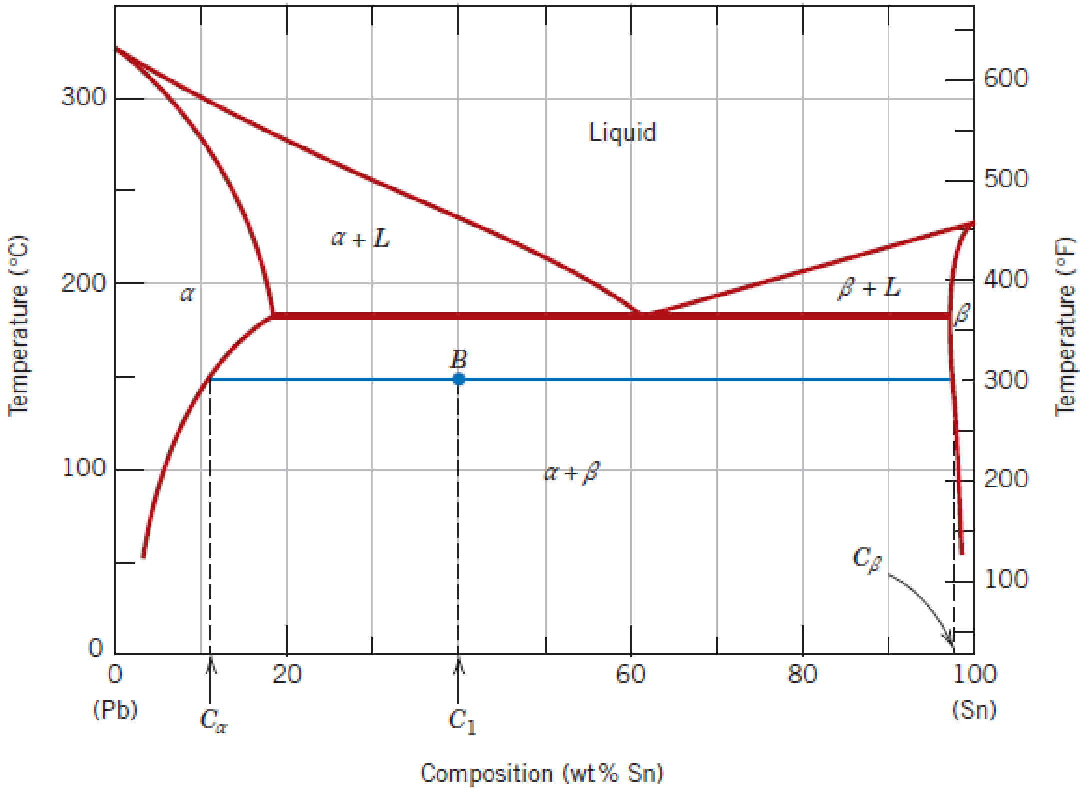
Determination of Phases Present and Computation of Phase Compositions

For a 40 wt% Sn–60 wt% Pb alloy at 150°C (300°F), (a) what phase(s) is (are) present?
(b) What is (are) the composition(s) of the phase(s)?

Solution

(a) Locate this temperature–composition point on the phase diagram (point *B* in Figure).
Inasmuch as it is within the *α* + *β* region, both *α* and *β* phases will coexist.

(b) Because two phases are present, it becomes necessary to construct a tie line across the *α* + *β* phase field at 150°C, as indicated in Figure. The composition of the *α* phase corresponds to the tie line intersection with the *α*/(*α* + *β*) solvus phase boundary—about 11 wt% Sn–89 wt% Pb, denoted as *Cα*. This is similar for the *β* phase, which has a composition of approximately 98 wt% Sn–2 wt% Pb (*Cβ*).



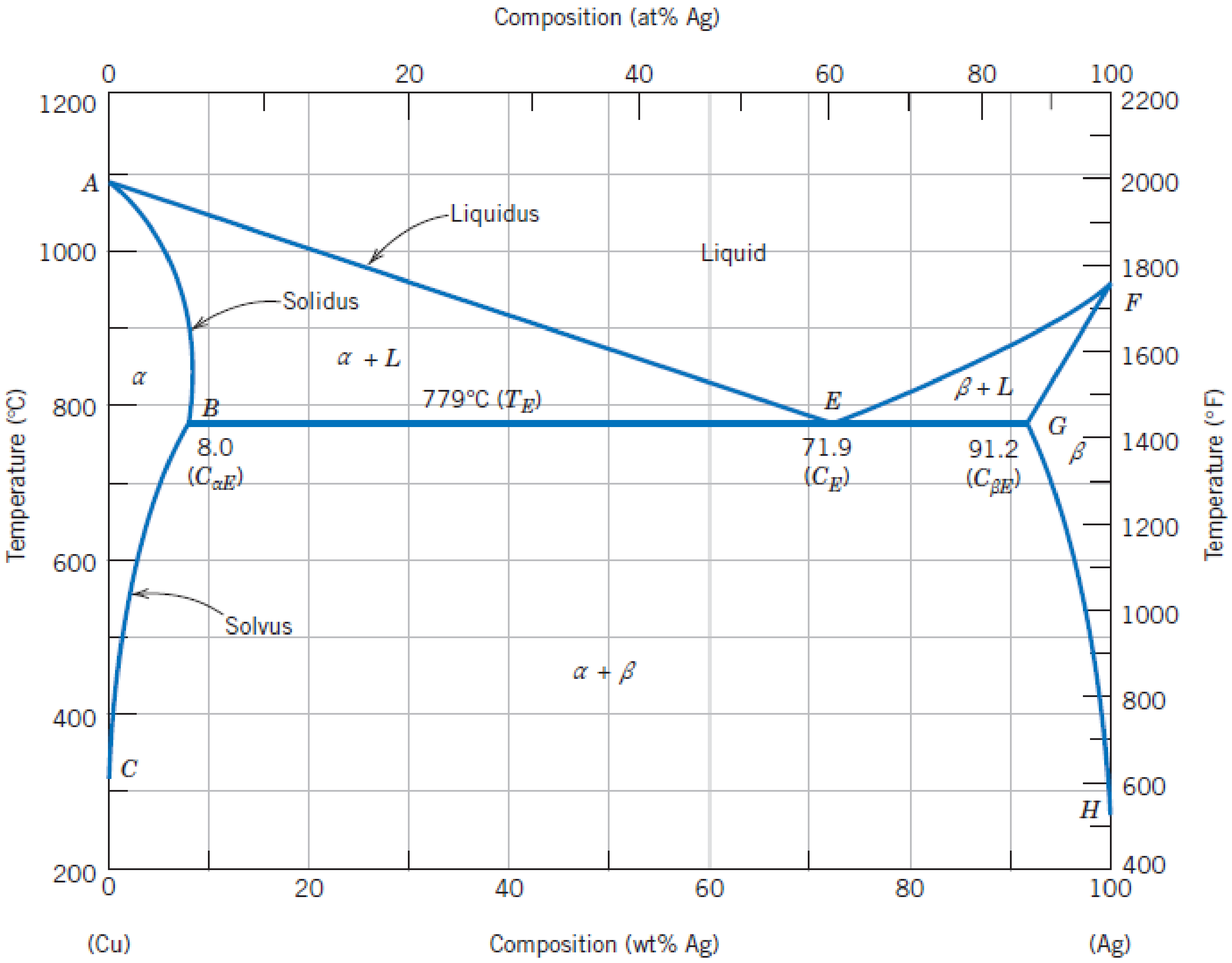
Practice example:

At 500℃ (930℉), what is the maximun solubility (a) of Cu in Ag? (b) Of Ag in Cu?

Solution

(a) From Figure 5, the maximum solubility of Cu in Ag at 500°C corresponds to the position of the β–(α + β) phase boundary at this temperature, or to about 2 wt% Cu.

(b) From this same figure, the maximum solubility of Ag in Cu corresponds to the position of the α–(α + β) phase boundary at this temperature, or about 1.5 wt% Ag.



Eutectoid reaction

- Upon cooling, a solid δ phase transforms into two other solid phases (γ and ε) according to the reaction The reverse reaction occurs upon heating. It is called a **eutectoid** (or eutectic-like) **reaction;** the eutectoid point is labeled *E* in Figure 6.

- For this reaction, 560°C is the eutectoid isotherm, 74 wt% Zn–26 wt% Cu is the eutectoid composition, whereas the γ and ε product phases have compositions of 70 wt% Zn–30 wt% Cu and 78 wt% Zn–22 wt% Cu, respectively. The feature distinguishing *eutectoid* from *eutectic* is that one solid phase instead of a liquid transforms into two other solid phases at a single temperature.

- A eutectoid reaction found in the iron–carbon system (Section 10.19) is very important in the heat treatment of steels.

$$\delta \underset{\text{heating}}{\overset{\text{cooling}}{\rightleftharpoons}} \gamma + \varepsilon$$

Peritectic reaction

The **peritectic reaction** is another invariant reaction involving three phases at equilibrium. With this reaction, upon heating, one solid phase transforms into a liquid phase and another solid phase. A peritectic exists for the copper-zinc system (Figure 6, point *P*) at 598°C (1108°F) and 78.6 wt% Zn–21.4 wt% Cu; this reaction is as follows:

$$\delta + L \underset{\text{heating}}{\overset{\text{cooling}}{\rightleftharpoons}} \varepsilon$$

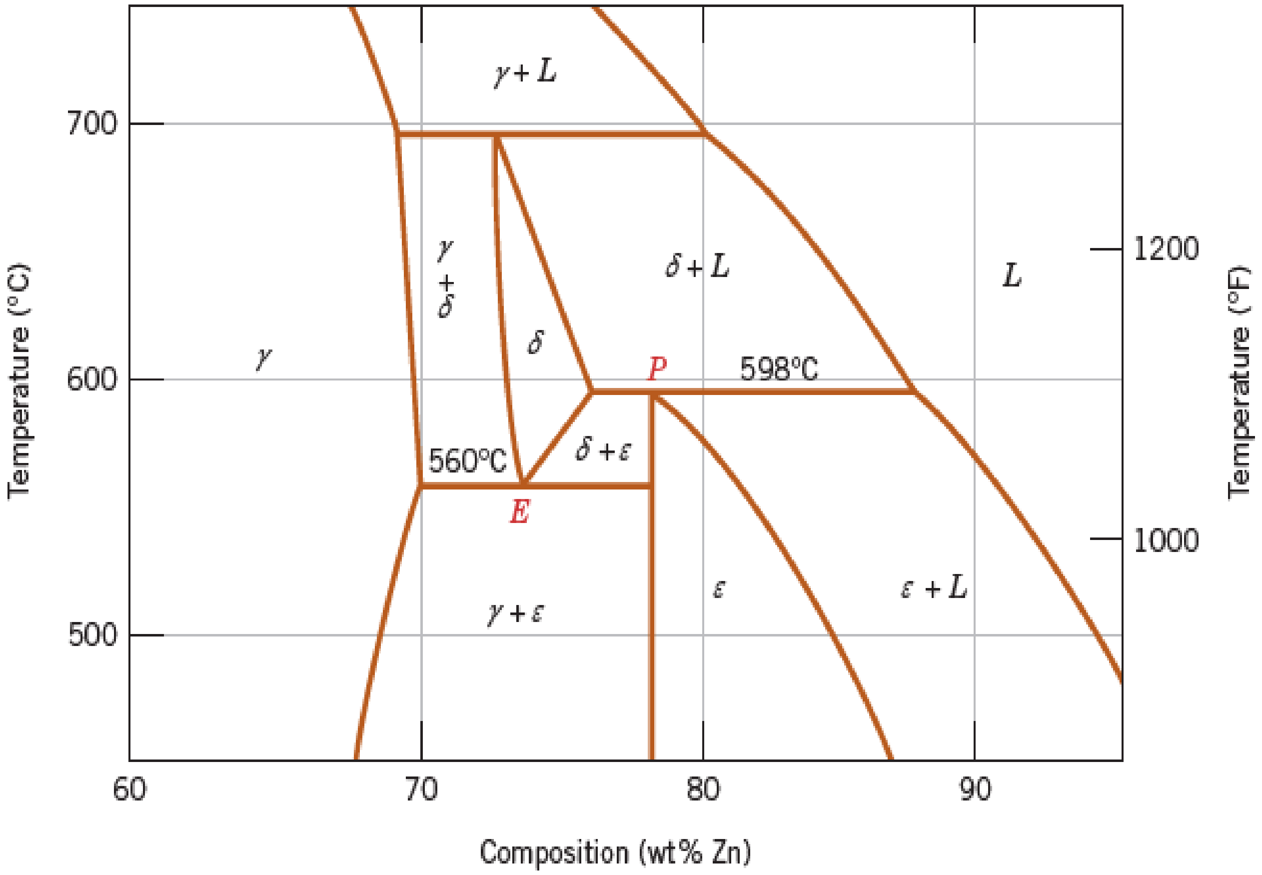


Figure 6 A region of the copper-zinc phase diagram that has been enlarged to show eutectoid and peritectic points, labeled *E* (560°C, 74 wt% Zn) and *P* (598°C, 78.6 wt% Zn), respectively.

TERNARY PHASE DIAGRAMS

- Phase diagrams have also been determined for metallic systems containing more than two components;
- However, their representation and interpretation can be exceedingly complex.
- For example, a ternary, or three-component, composition–temperature phase diagram in its entirety is depicted by a three-dimensional model.

THE GIBBS PHASE RULE

- The construction of phase diagrams—as well as some of the principles governing the conditions for phase equilibria—are dictated by laws of thermodynamics. One of these is the **Gibbs phase rule,** *proposed by the 19th century physicist J. Willard Gibbs.*
- This rule represents a criterion for the number of phases that coexist within a system at equilibrium and is expressed by the simple equation

$$P + F = C + N$$

where

- *P* is the number of phases present.
- The parameter *F* is termed the *number of degrees of freedom,* or the number of externally controlled variables (e.g., temperature, pressure, composition) that must be specified to define the state of the system completely.
- Expressed another way, *F* is the number of these variables that can be changed independently without altering the number of phases that coexist at equilibrium.
- The parameter *C* represents the number of components in the system. Components are normally elements or stable compounds and, in the case of phase diagrams, are the materials at the two extremes of the horizontal compositional axis (e.g., H_2O and $C_{12}H_{22}O_{11}$, and Cu and Ni, for the phase diagrams).
- Finally, *N* is the number of noncompositional variables (e.g., temperature and pressure).

$$\boxed{P + F = C + N}$$

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Example:

$P + F = C + N$ Cu Ag

The phase rule by applying it to binary temperature–composition phase diagrams, specifically the copper–silver system, Because pressure is constant (1 atm), the parameter N is 1—temperature is the only noncompositional variable. Then:

$$P + F = C + 1$$

The number of components C is 2 (namely, Cu and Ag), and

$$P + F = 2 + 1 = 3$$

or

$$F = 3 - P$$

Consider the case of single-phase fields on the phase diagram (e.g., α, β, and liquid regions). Because only one phase is present, $P = 1$ and

$$F = 3 - P = 3 - 1 = 2$$

$F = 2$

This means that to completely describe the characteristics of any alloy that exists within one of these phase fields, we must specify two parameters—composition and temperature, which locate, respectively, the horizontal and vertical positions of the alloy on the phase diagram.

THE IRON–IRON CARBIDE ($Fe–Fe_3C$) PHASE DIAGRAM

- At room temperature, the stable form, called **ferrite,** or α-iron, has a BCC crystal structure.
- Ferrite experiences a polymorphic transformation to FCC **austenite,** or γ-iron, at 912°C (1674°F).
- This austenite persists to 1394°C (2541°F), at which temperature the FCC austenite reverts back to a BCC phase known as δ-ferrite, which finally melts at 1538°C (2800°F).
- The composition axis in Figure 7 extends only to 6.70 wt% C; at this concentration the intermediate compound iron carbide, or **cementite** (Fe_3C), is formed, which is represented by a vertical line on the phase diagram.

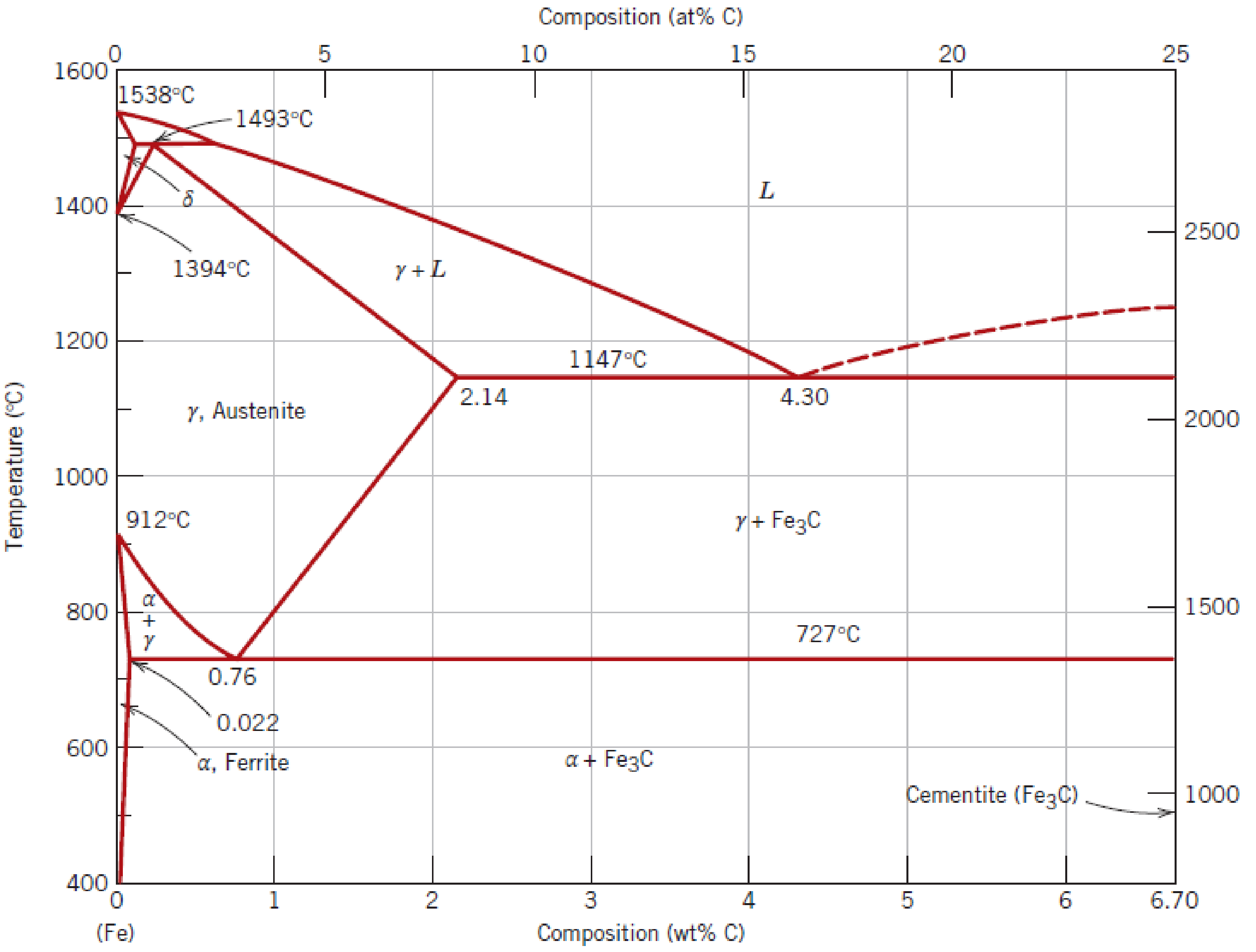


Figure 7 The iron–iron carbide phase diagram.

- Thus, the iron–carbon system may be divided into two parts: an iron-rich portion, as in Figure 7, and the other (not shown) for compositions between 6.70 and 100 wt% C (pure graphite).
- In practice, all steels and cast irons have carbon contents less than 6.70 wt% C; therefore, we consider only the iron–iron carbide system.

- The two-phase regions are labeled in Figure 7. It may be noted that one eutectic exists for the iron–iron carbide system, at 4.30 wt% C and 1147°C (2097°F); for this eutectic reaction,

$$L \underset{\text{heating}}{\overset{\text{cooling}}{\rightleftharpoons}} \gamma + Fe_3C$$

the liquid solidifies to form austenite and cementite phases. Subsequent cooling to room temperature promotes additional phase changes.

- It may be noted that a eutectoid invariant reaction exists at a composition of 0.76 wt% C and a temperature of 727°C (1341°F). This eutectoid reaction may be represented by

$$\gamma(0.76 \text{ wt\% C}) \underset{\text{heating}}{\overset{\text{cooling}}{\rightleftharpoons}} \alpha(0.022 \text{ wt\% C}) + Fe_3C(6.70 \text{ wt\% C})$$

or, upon cooling, the solid γ phase is transformed into α-iron and cementite. (Eutectoid phase transformations).

DEVELOPMENT OF MICROSTRUCTURE IN IRON–CARBON ALLOYS

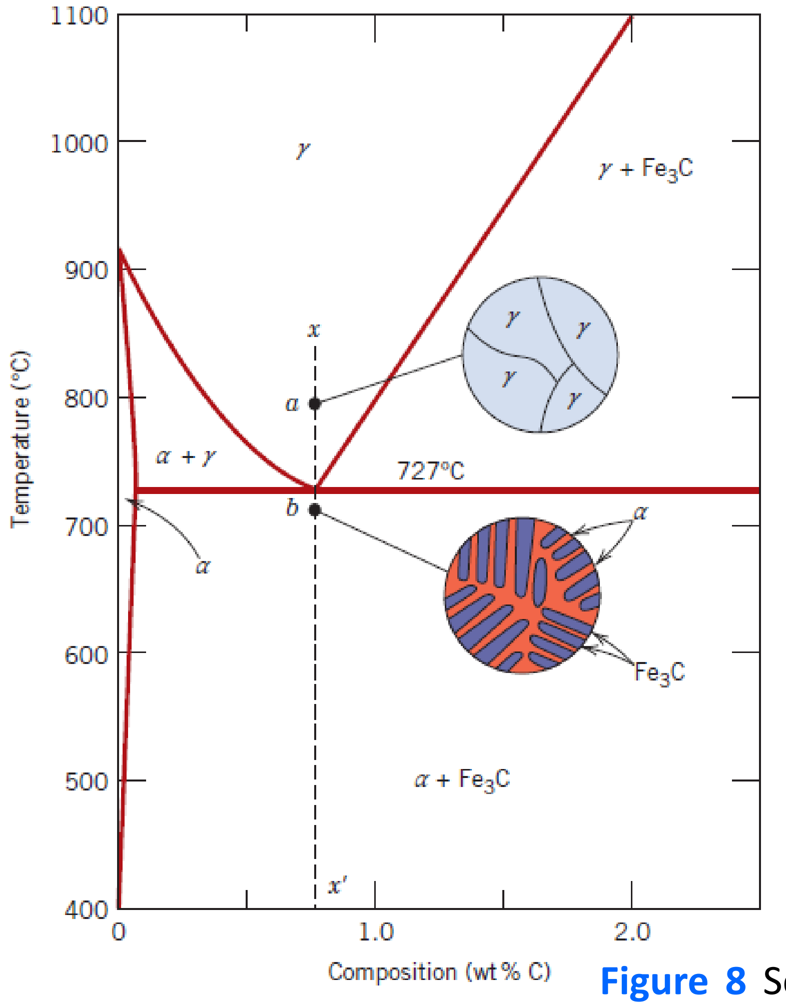


Figure 8 Schematic representations of the microstructures for an iron–carbon alloy of eutectoid composition (0.76 wt% C) above and below the eutectoid temperature.

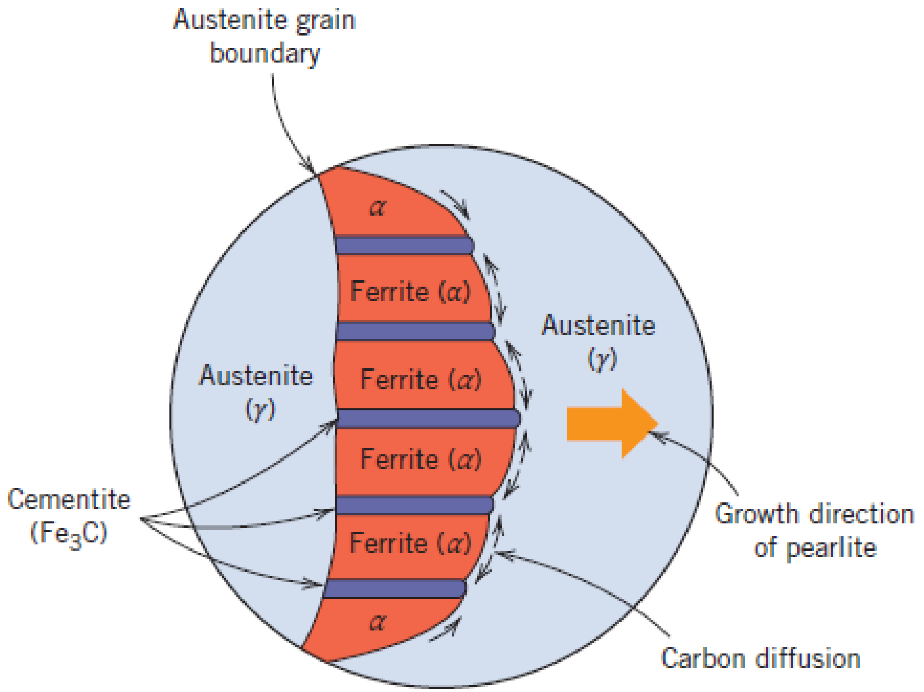


Figure 9 Schematic representation of the formation of ***pearlite*** from austenite; the direction of carbon diffusion is indicated by arrows.

Hypoeutectoid Alloys

- Consider a composition *C*0 to the left of the eutectoid, between 0.022 and 0.76 wt% C; this is termed a **hypoeutectoid** ("less than eutectoid") **alloy.**

- The ferrite present in the pearlite is called *eutectoid ferrite,* whereas the other, which formed above T_e, is termed **proeutectoid** (meaning "pre- or before eutectoid") **ferrite.**

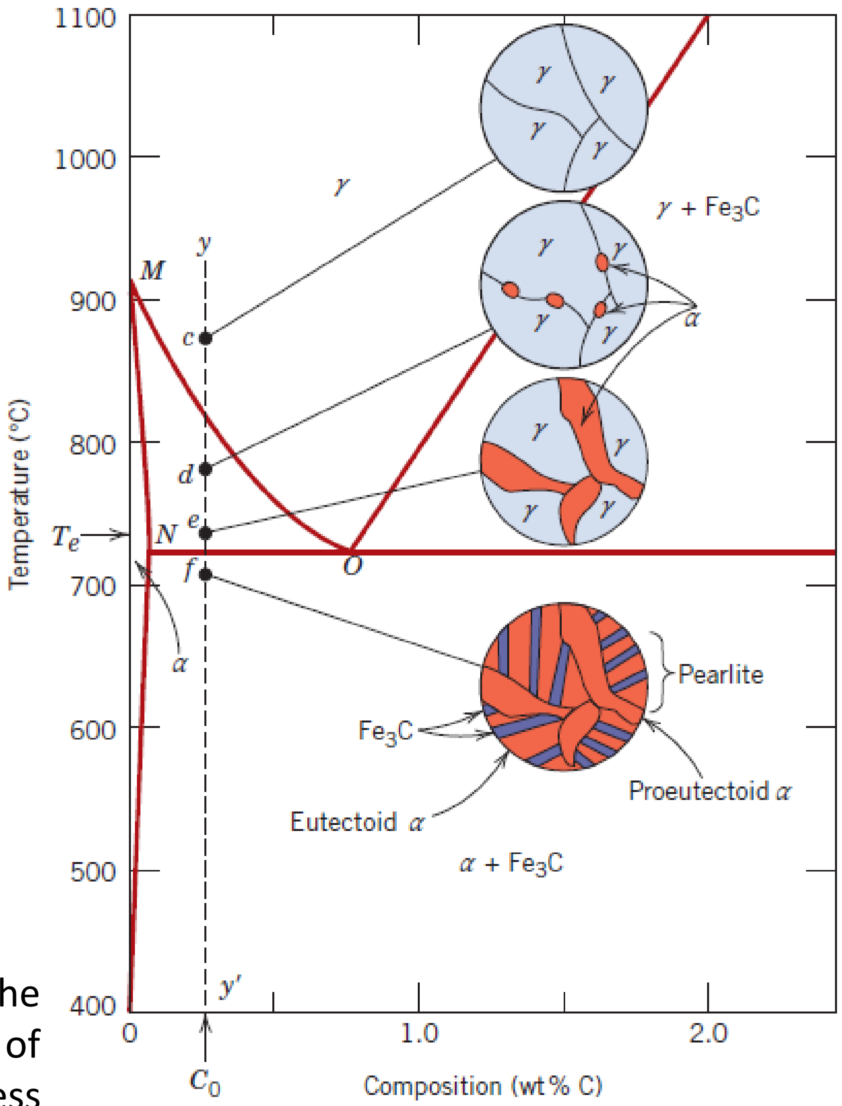


Figure 10 Schematic representations of the microstructures for an iron– carbon alloy of hypoeutectoid composition C_0 (containing less than 0.76 wt% C) as it is cooled from within the austenite phase region to below the eutectoid temperature.

Hypereutectoid Alloys

- Analogous transformations and microstructures result for **hypereutectoid alloys**—those containing between 0.76 and 2.14 wt% C—that are cooled from temperatures within the γ-phase field. Consider an alloy of composition C_1 in Figure 11 that, upon cooling, moves down the line *zz*′.

- Upon cooling into the γ + Fe3C phase field—say, to point *h*—the cementite phase begins to form along the initial γ grain boundaries, similar to the α phase in Figure 11, point *d*. This cementite is called **proeutectoid cementite**—that which forms before the eutectoid reaction.

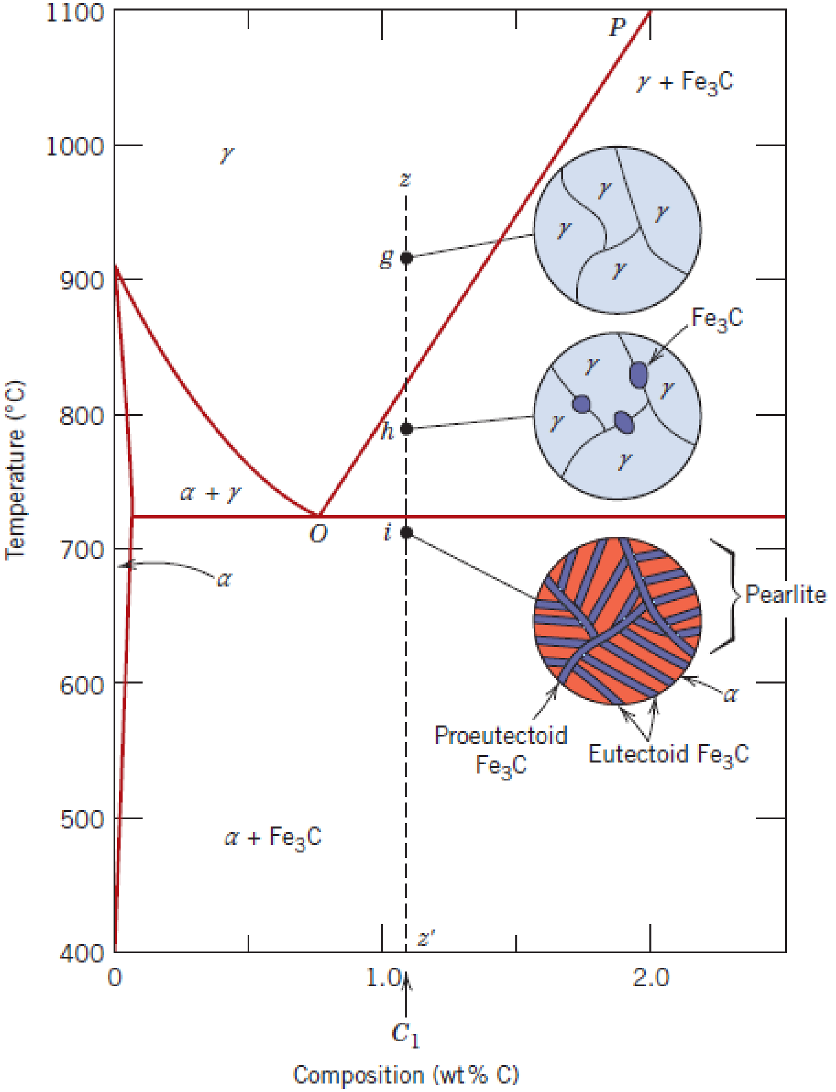


Figure 11 Schematic representations of the microstructures for an iron–carbon alloy of hypereutectoid composition C_1 (containing between 0.76 and 2.14 wt% C) as it is cooled from within the austenite-phase region to below the eutectoid temperature.

Nonequilibrium Cooling

- The microstructural development of iron–carbon alloys it has been assumed that, upon cooling, conditions of metastable equilibrium have been continuously maintained; that is, sufficient time has been allowed at each new temperature for any necessary adjustment in phase compositions and relative amounts as predicted from the Fe–Fe_3C phase diagram.

- In most situations these cooling rates are impractically slow and unnecessary; in fact, on many occasions nonequilibrium conditions are desirable.

- Two nonequilibrium effects of practical importance are

  1. the occurrence of phase changes or transformations at temperatures other than those predicted by phase boundary lines on the phase diagram, and

  2. The existence at room temperature of nonequilibrium phases that do not appear on the phase diagram.

Mechanisms of Strengthening in Metals

- Strengthening mechanisms is the relation between dislocation motion and mechanical behavior of metals.

- Because macroscopic plastic deformation corresponds to the **motion of large numbers of dislocations**, ***the ability of a metal to deform plastically depends on the ability of dislocations to move***.

- Because hardness and strength (both yield and tensile) are related to the ease with which plastic deformation can be made to occur, by **reducing the mobility of dislocations**, the mechanical strength may be enhanced—that is, greater mechanical forces are required to initiate plastic deformation.

- In contrast, the more unconstrained the dislocation motion, the greater the facility with which a metal may deform, and the softer and weaker it becomes. Virtually all strengthening techniques rely on this simple principle: *Restricting or hindering dislocation motion renders a material harder and stronger*.

STRENGTHENING BY GRAIN SIZE REDUCTION

During plastic deformation, slip or dislocation motion must take place across this common boundary—say, from grain A to grain B in Figure 12. The grain boundary acts as a barrier to dislocation motion for two reasons:

1. Because the two grains are of different orientations, a dislocation passing into grain B must change its direction of motion; this becomes more difficult as the crystallographic misorientation increases.

2. The atomic disorder within a grain boundary region results in a discontinuity of slip planes from one grain into the other.

- For high-angle grain boundaries, it may not be the case that dislocations traverse grain boundaries during deformation; rather, dislocations tend to "pile up" (or back up) at grain boundaries. These pileups introduce stress concentrations ahead of their slip planes, which generate new dislocations in adjacent grains.

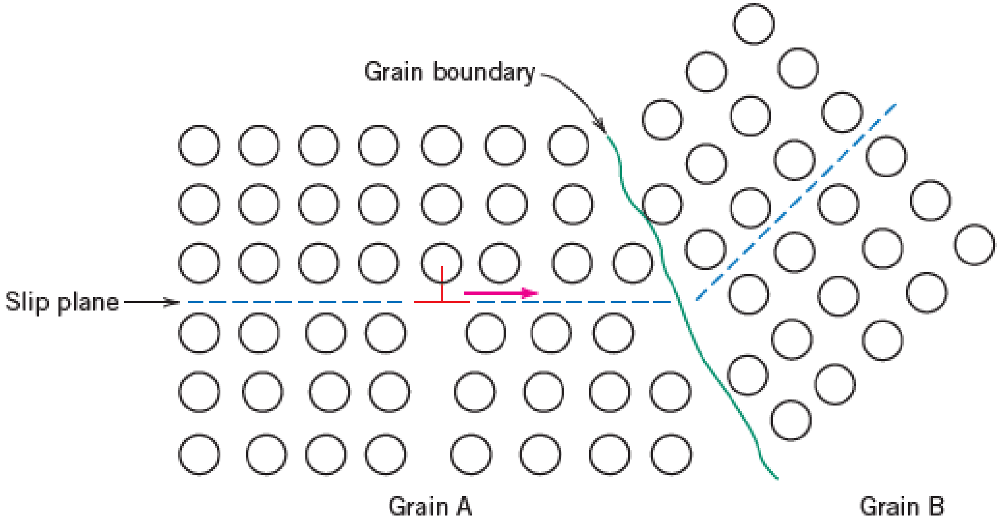


Figure 12 The motion of a dislocation as it encounters a grain boundary, illustrating how the boundary acts as a barrier to continued slip. Slip planes are discontinuous and change directions across the boundary.

- A fine-grained material (one that has small grains) is harder and stronger than one that is coarse grained because the former has a greater total grain boundary area to impede dislocation motion. For many materials, the yield strength σ_y varies with grain size according to

$$\sigma_y = \sigma_0 + k_y d^{-1/2}$$

- The ***Hall–Petch equation, d*** is the **average grain diameter**, and $\boldsymbol{\sigma_0}$ and $\boldsymbol{k_y}$ are **constants** for a particular material.

- Above equation is **not valid** for **both very large** (i.e., coarse) grain and **extremely fine grain** polycrystalline materials.

- *It should also be mentioned that grain size reduction improves not only the strength, but also the toughness of many alloys.*

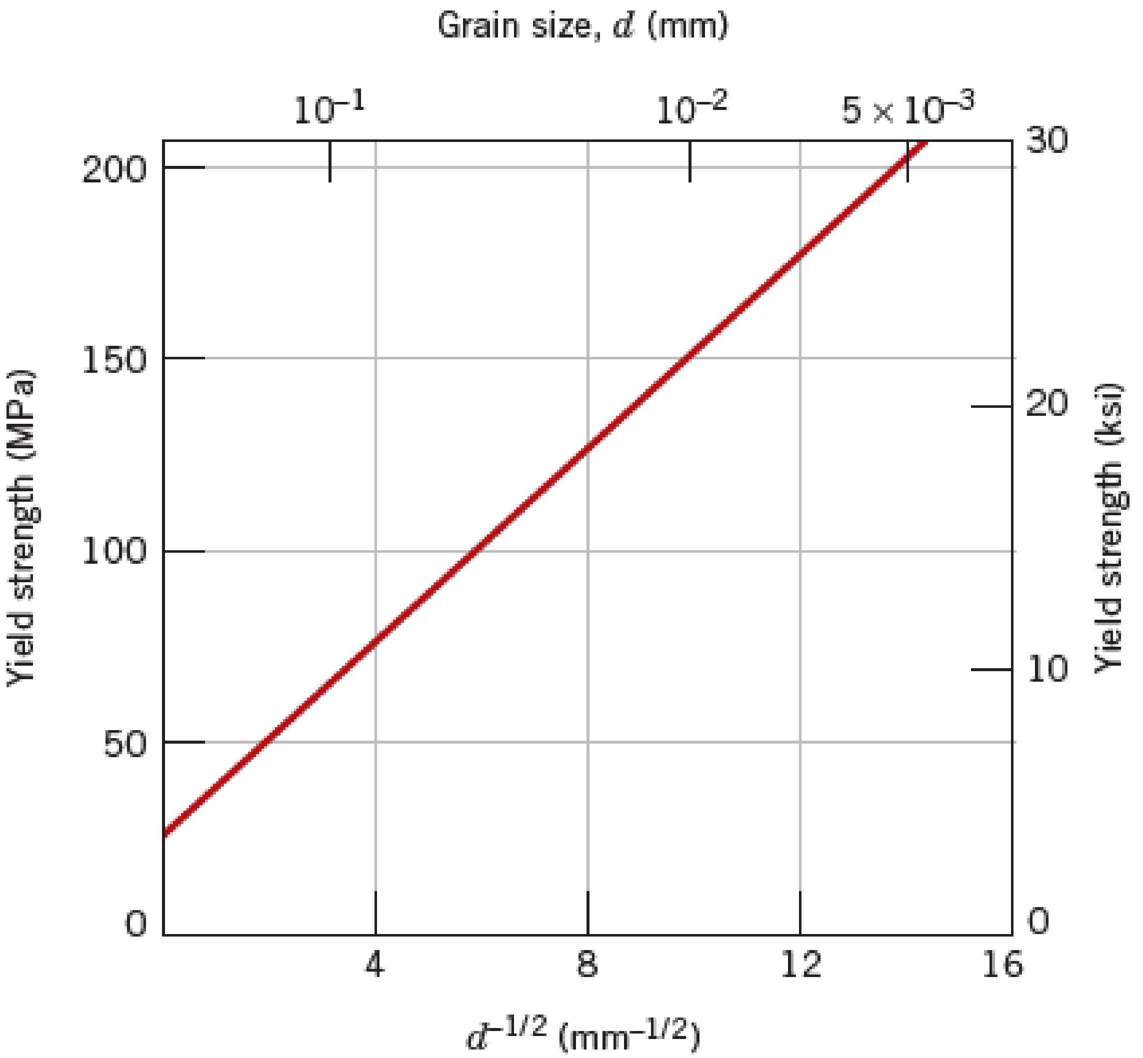


Figure 13 The influence of grain size on the yield strength of a 70 Cu–30 Zn brass alloy. Note that the grain diameter increases from right to left and is not linear.

SOLID-SOLUTION STRENGTHENING

- Another technique to strengthen and harden metals is alloying with impurity atoms that go into either substitutional or interstitial solid solution which is called **solid-solution strengthening.**

- High-purity metals are almost always softer and weaker than alloys composed of the same base metal.

- Increasing the concentration of the impurity results in an attendant increase in tensile and yield strengths, as indicated in Figures 14*a* and 14*b*, respectively, for nickel in copper; the dependence of ductility on nickel concentration is presented in Figure 14*c*.

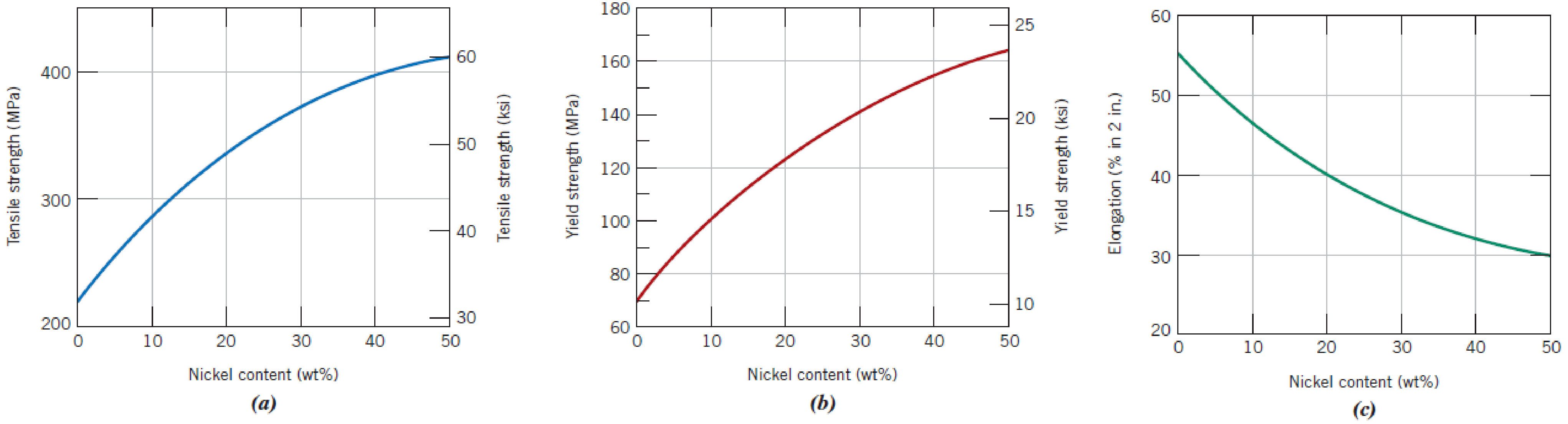


Figure 14 Variation with a nickel content of (*a*) tensile strength, (*b*) yield strength, and (*c*) ductility (%EL) for copper–nickel alloys, showing strengthening.

- Alloys are **stronger** than **pure metals** because impurity atoms that go into solid solutions typically **impose lattice strains** on the surrounding host atoms. Lattice strain field interactions between dislocations and these impurity atoms result, and, consequently, **dislocation movement is restricted**.
- These solute atoms tend to diffuse and segregate around dislocations in such a way as to reduce the overall strain energy, to cancel some of the strain in the lattice surrounding a dislocation.
- To accomplish this, a smaller impurity atom is located where its tensile strain partially nullifies some of the dislocation's compressive strain.

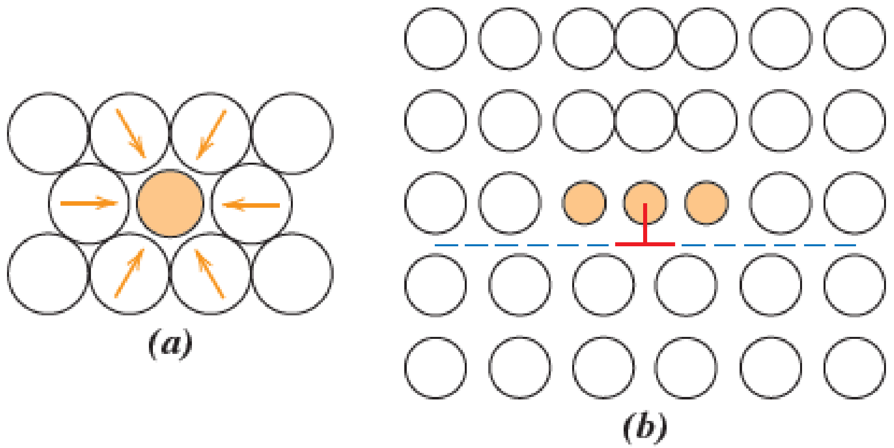


Figure 15 (*a*) Representation of tensile lattice strains imposed on host atoms by a smaller substitutional impurity atom. (*b*) Possible locations of smaller impurity atoms relative to an edge dislocation such that there is partial cancellation of impurity–dislocation lattice strains.

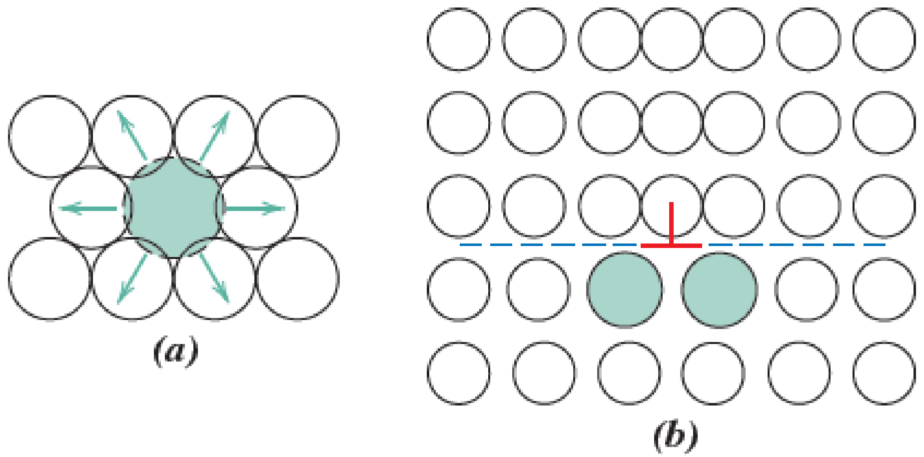


Figure 16 (*a*) Representation of compressive strains imposed on host atoms by a larger substitutional impurity atom. (*b*) Possible locations of larger impurity atoms relative to an edge dislocation such that there is partial cancellation of impurity–dislocation lattice strains.

STRAIN HARDENING

- **Strain hardening** is the phenomenon by which a ductile metal becomes harder and stronger as it is plastically deformed.

- Sometimes it is also called *work hardening* or, because the temperature at which deformation takes place is "cold" relative to the absolute melting temperature of the metal, **cold working.** Most metals strain harden at room temperature.

- It is sometimes convenient to express the degree of plastic deformation as *percent cold work* rather than as strain. Percent cold work (%CW) is defined as

$$\%CW = \left(\frac{A_0 - A_d}{A_0}\right) \times 100$$

where A_0 is the original area of the cross-section that experiences deformation and A_d is the area after deformation.

- The strain-hardening phenomenon is explained on the basis of dislocation–dislocation strain field interactions.
- The dislocation density in a metal increases with deformation or cold work because of dislocation multiplication or the formation of new dislocations.
- Consequently, the average distance of separation between dislocations decreases—the dislocations are positioned closer together. On average, dislocation–dislocation strain interactions are repulsive. The net result is that the motion of a dislocation is hindered by the presence of other dislocations.
- As the dislocation density increases, this resistance to dislocation motion by other dislocations becomes more pronounced.
- Thus, the imposed stress necessary to deform a metal increases with increasing cold work.
- Strain hardening is often utilized commercially to enhance the mechanical properties of metals during fabrication procedures. The effects of strain hardening may be removed by an annealing heat treatment.

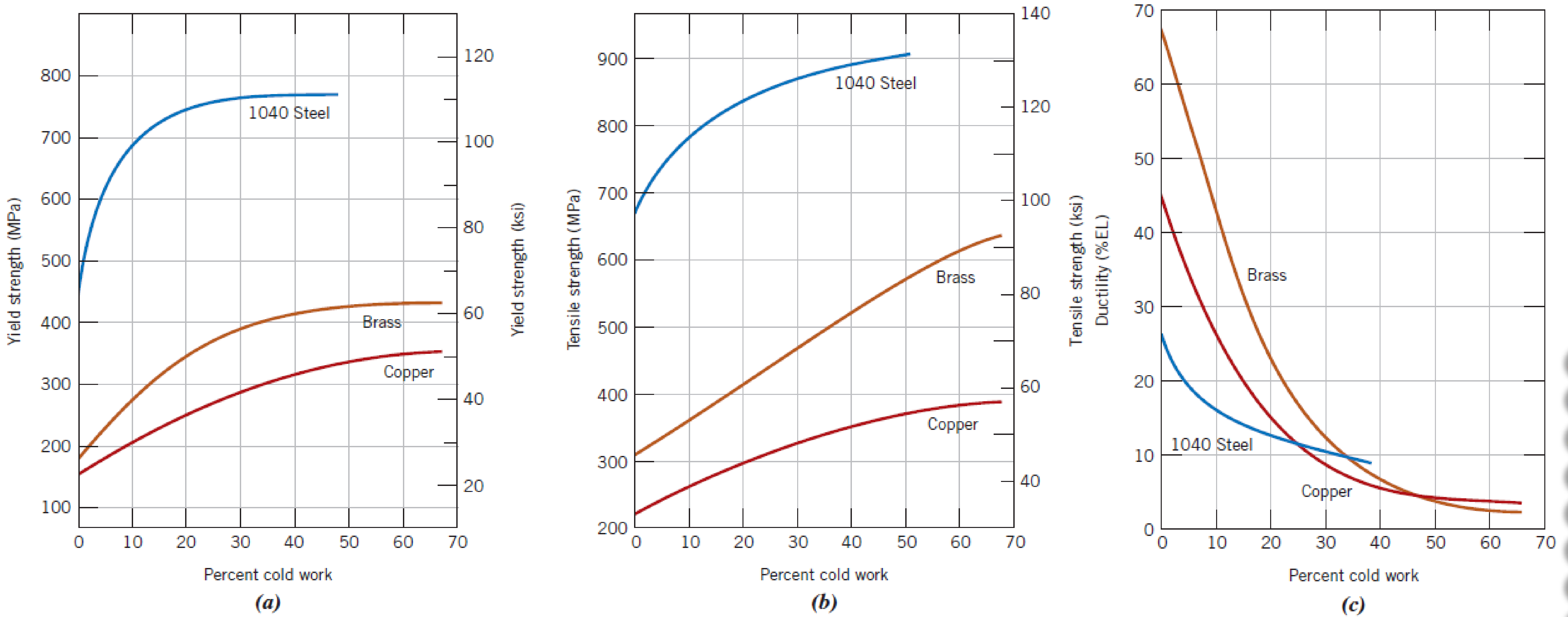


Figure 17 For 1040 steel, brass, and copper, (*a*) the increase in yield strength, (*b*) the increase in tensile strength, and (*c*) the decrease in ductility (%EL) with percent cold work.

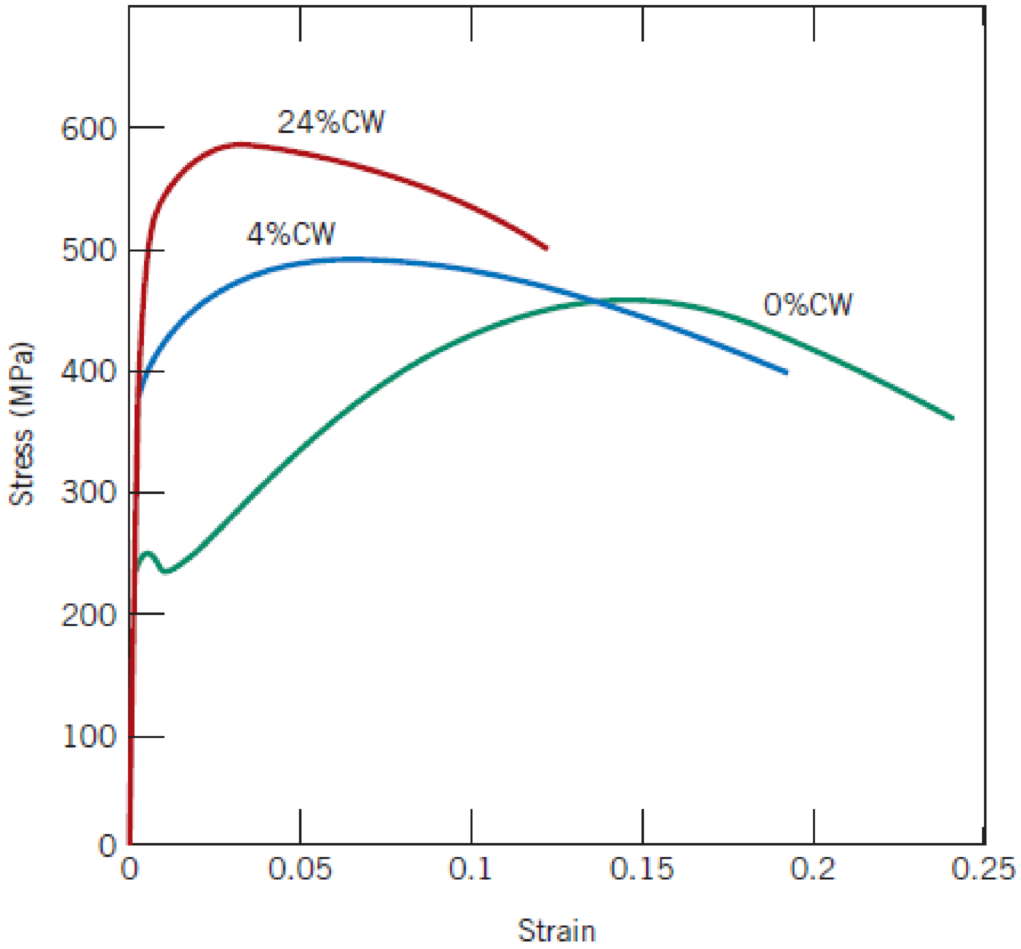


Figure 18 The influence of cold work on the stress–strain behavior of low-carbon steel; curves are shown for 0%CW, 4%CW, and 24%CW.

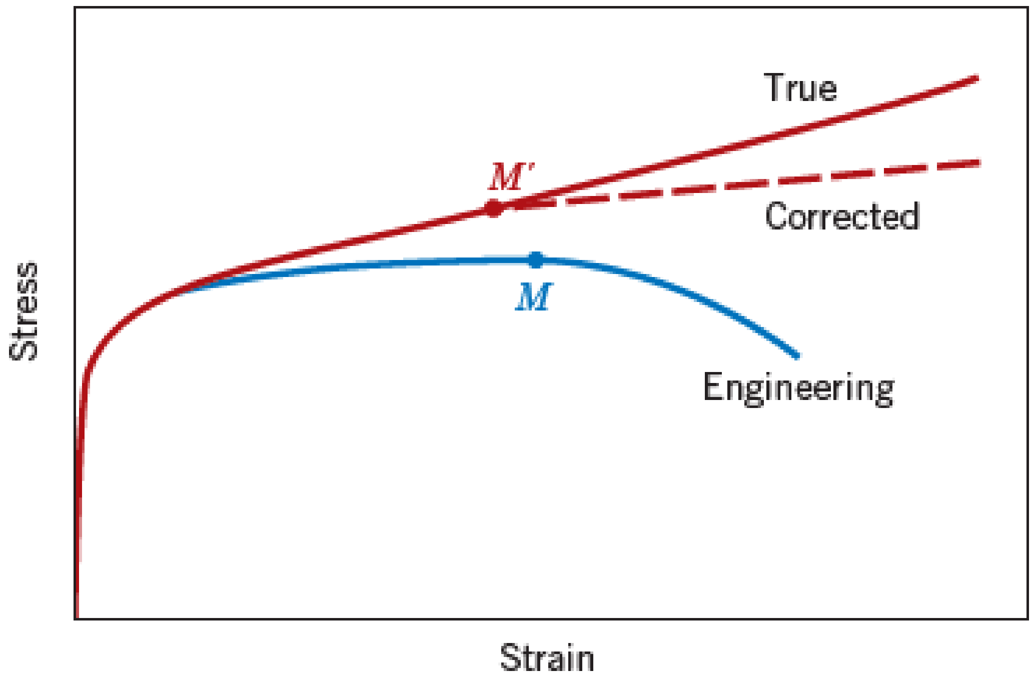


Figure 19 A comparison of typical tensile engineering stress–strain and true stress–strain behaviors. Necking begins at point *M* on the engineering curve, which corresponds to *M*′ on the true curve. The "corrected" true stress–strain curve takes into account the complex stress state within the neck region.

EXAMPLE PROBLEM

1. Compute the tensile strength and ductility (%EL) of a cylindrical copper rod if it is cold-worked such that the diameter is reduced from 15.2 mm to 12.2 mm.

$$\%\mathrm{CW} = \frac{\left(\frac{15.2\ \mathrm{mm}}{2}\right)^2 \pi - \left(\frac{12.2\ \mathrm{mm}}{2}\right)^2 \pi}{\left(\frac{15.2\ \mathrm{mm}}{2}\right)^2 \pi} \times 100 = 35.6\%$$

The tensile strength is read directly from the curve for copper (Figure 17b) as 340 Mpa. From Figure 17c, the ductility at 35.6%CW is about 7%EL.

Recovery, Recrystallization, and Grain Growth

- Plastically deforming a polycrystalline metal specimen at temperatures that are low relative to its absolute melting temperature produces microstructural and property changes that include:

  1. a change in grain shape,
  2. strain hardening and
  3. an increase in dislocation density

- Some fraction of the energy expended in deformation is stored in the metal as strain energy, which is associated with tensile, compressive, and shear zones around the newly created dislocations.

- Furthermore, other properties, such as electrical conductivity and corrosion resistance, may be modified as a consequence of plastic deformation.

- These properties and structures may revert back to the pre–pre-cold-worked states by appropriate heat treatment (sometimes termed an annealing treatment).

- Such restoration results from two different processes that occur at elevated temperatures: recovery and recrystallization, which may be followed by grain growth.

RECOVERY

- During **recovery,** some of the stored internal strain energy is relieved by virtue of dislocation motion (in the absence of an externally applied stress), as a result of enhanced atomic diffusion at the elevated temperature.

- There is some reduction in the number of dislocations, and dislocation configurations (similar to that shown in Figure 20) are produced having low strain energies. In addition, physical properties such as electrical and thermal conductivities recover to their pre-cold-worked states.

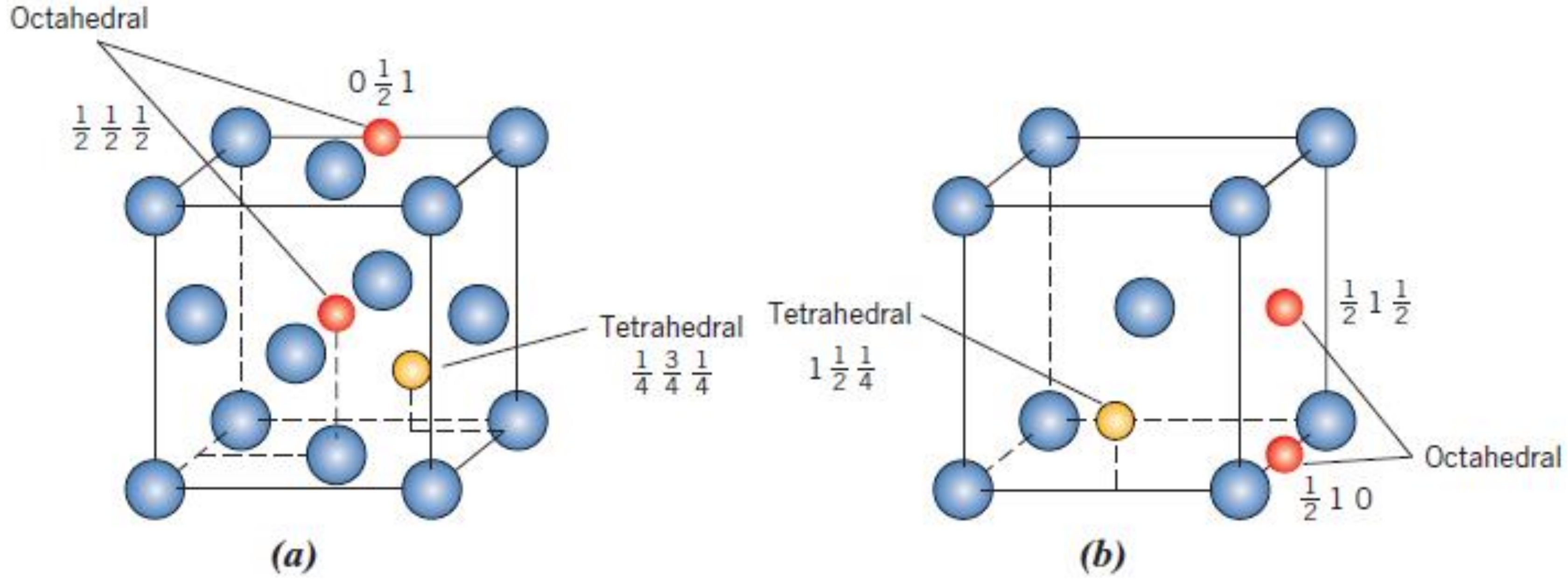


Figure 20 Locations of tetrahedral and octahedral interstitial sites within (*a*) FCC and (*b*) BCC unit cells.

RECRYSTALLIZATION

- Even after recovery is complete, the grains are still in a relatively high strain energy state.
- **Recrystallization** is the formation of a new set of strain-free and equiaxed grains (i.e., having approximately equal dimensions in all directions) that have low dislocation densities and are characteristic of the pre-cold-worked condition.
- The driving force to produce this new grain structure is the difference in internal energy between the strained and unstrained material.
- The new grains form as very small nuclei and grow until they completely consume the parent material, processes that involve short-range diffusion.

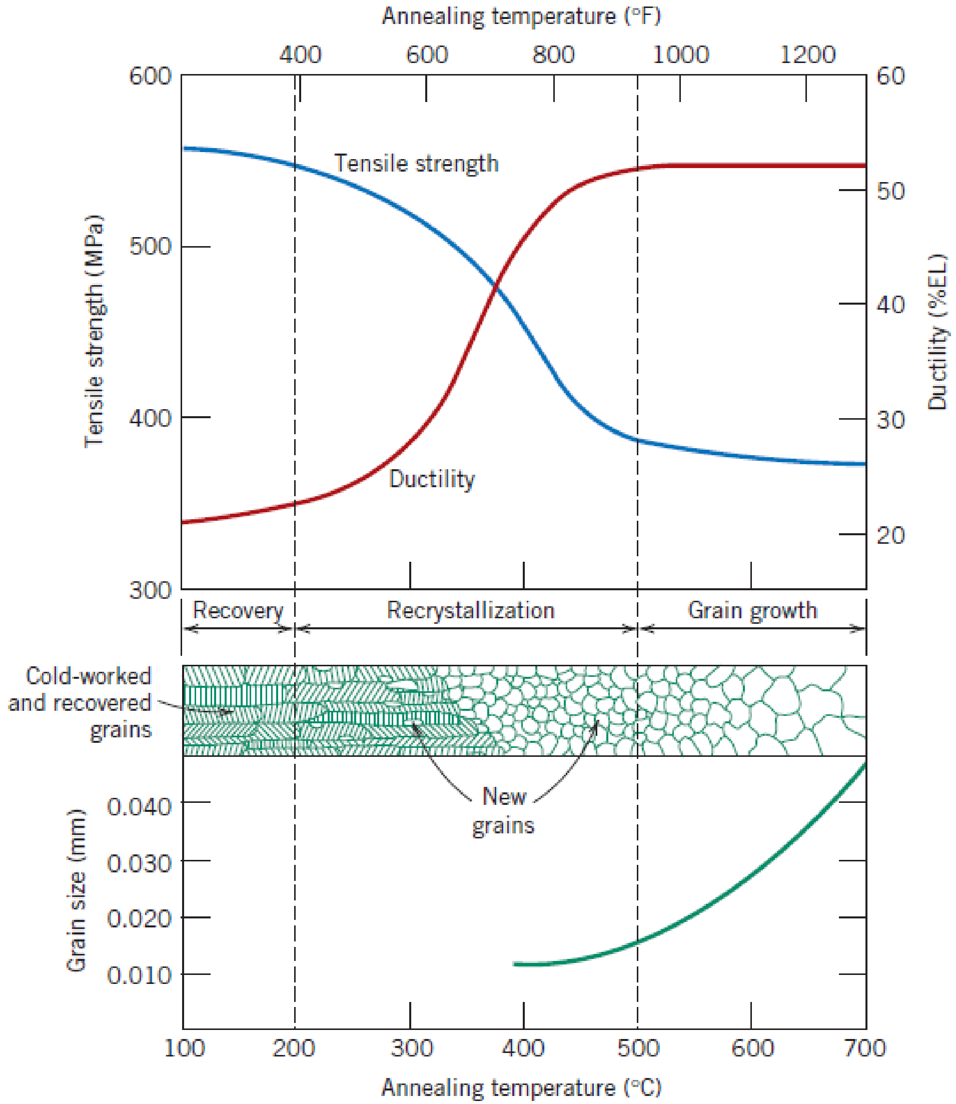


Figure 21 The influence of annealing temperature (for an annealing time of 1 h) on the tensile strength and ductility of a brass alloy.

- The recrystallization behavior of a particular metal alloy is sometimes specified in terms of a **recrystallization temperature,** the temperature at which recrystallization just reaches completion in 1 h.

- Thus, the recrystallization temperature for the brass alloy of Figure 21 is about 450°C (850°F).

- Typically, it is between one-third and one-half of the absolute melting temperature of a metal or alloy and depends on several factors, including the amount of prior cold work and the purity of the alloy.

- Increasing the percent of cold work enhances the rate of recrystallization, with the result that the recrystallization temperature is lowered, and it approaches a constant or limiting value at high deformations.

GRAIN GROWTH

- After recrystallization is complete, the strain-free grains will continue to grow if the metal specimen is left at the elevated temperature, this phenomenon is called **grain growth.**

- Grain growth does not need to be preceded by recovery and recrystallization; it may occur in all polycrystalline materials—metals and ceramics alike.

- An energy is associated with grain boundaries, As grains increase in size, the total boundary area decreases, yielding an attendant reduction in the total energy; this is the driving force for grain growth.

- Grain growth occurs by the migration of grain boundaries. Obviously, not all grains can enlarge, but large ones grow at the expense of small ones that shrink.

- Thus, the average grain size increases with time, and at any particular instant, there exists a range of grain sizes. Boundary motion is just the short-range diffusion of atoms from one side of the boundary to the other (Figure 22).

- For many polycrystalline materials, the grain diameter d varies with time *t* according to the relationship

$$d^n - d_0^n = Kt$$

where d_0 is the initial grain diameter at $t = 0$, and K and n are time-independent constants; the value of n is generally equal to or greater than 2.

- At lower temperatures the curves are linear. Furthermore, grain growth proceeds more rapidly as temperature increases—that is, the curves are displaced upward to larger grain sizes. This is explained by the enhancement of diffusion rate with rising temperature (Figure 23).

- The mechanical properties at room temperature of a fine-grained metal are usually superior (i.e., higher strength and toughness) to those of coarse-grained ones.

- If the grain structure of a single-phase alloy is coarser than that desired, refinement may be accomplished by plastically deforming the material, then subjecting it to a recrystallization heat treatment, as described previously.

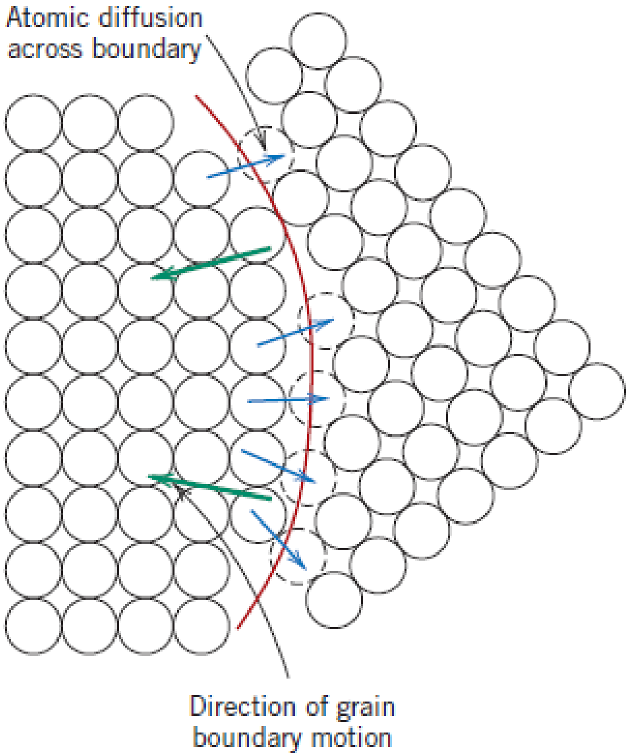


Figure 22 Schematic representation of grain growth via atomic diffusion.

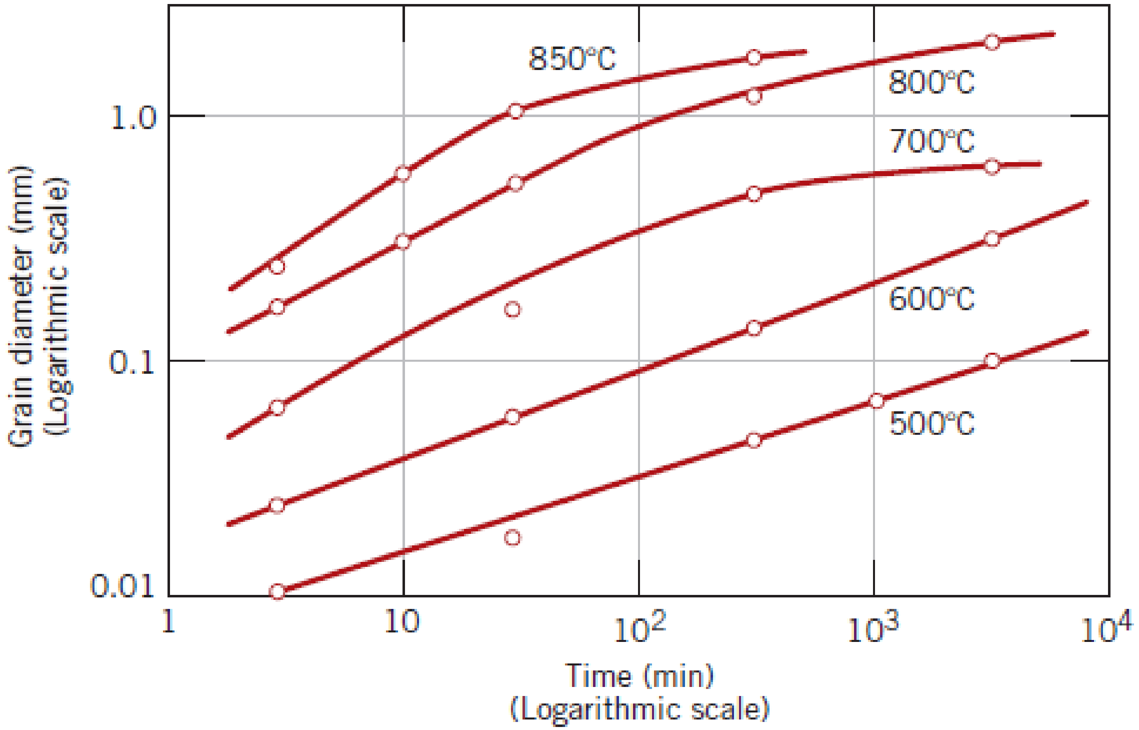


Figure 23 The logarithm of grain diameter versus the logarithm of time for grain growth in brass at several temperatures.

EXAMPLE PROBLEM

When a hypothetical metal having a grain diameter of 8.2×10^{-3} mm is heated to 500°C for 12.5 min, the grain diameter increases to 2.7×10^{-2} mm. Compute the grain diameter when a specimen of the original material is heated at 500°C for 100 min. Assume the grain diameter exponent *n* has a value of 2.

$$d^2 - d_0^2 = Kt$$

This leads to

$$K = \frac{(2.7 \times 10^{-2}\ \text{mm})^2 - (8.2 \times 10^{-3}\ \text{mm})^2}{12.5\ \text{min}}$$
$$= 5.29 \times 10^{-5}\ \text{mm}^2/\text{min}$$

To determine the grain diameter after a heat treatment at 500°C for 100 min, we must manipulate Equation 8.10 such that d becomes the dependent variable—that is,

$$d = \sqrt{d_0^2 + Kt}$$

Substitution into this expression of $t = 100$ min, as well as values for d_0 and K, yields

$$d = \sqrt{(8.2 \times 10^{-3}\ \text{mm})^2 + (5.29 \times 10^{-5}\ \text{mm}^2/\text{min})(100\ \text{min})}$$
$$= 0.0732\ \text{mm}$$

FUNDAMENTALS OF FRACTURE

A simple fracture is the separation of a body into two or more pieces in response to an imposed stress that is static (i.e., constant or slowly changing with time) and at temperatures that are low relative to the melting temperature of the material.

DUCTILE FRACTURE

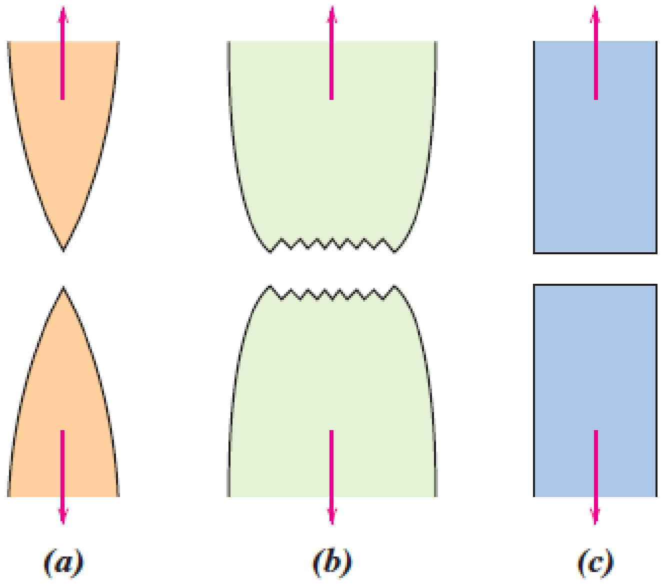


Figure 24 (*a*) Highly ductile fracture in which the specimen necks down to a point. (*b*) Moderately ductile fracture after some necking. (*c*) Brittle fracture without any plastic deformation.

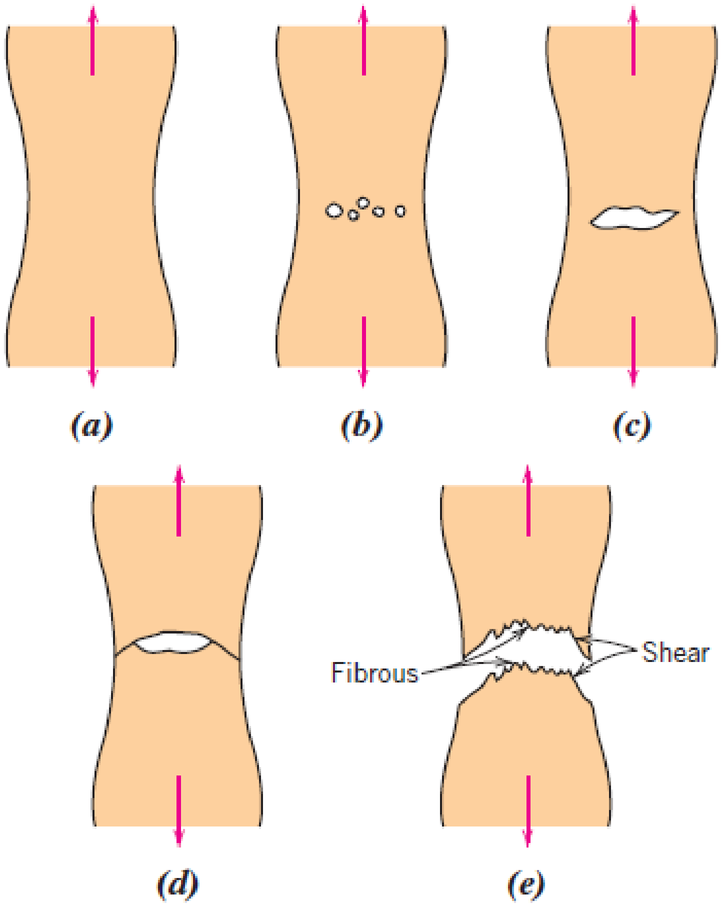


Figure 25 Stages in the cup-and-cone fracture. (*a*) Initial necking. (*b*) Small cavity formation. (*c*) Coalescence of cavities to form a crack. (*d*) Crack propagation. (*e*) Final shear fracture at a 45° angle relative to the tensile direction.

Fractographic Studies: detailed information regarding the mechanism of fracture under SEM

BRITTLE FRACTURE

- The direction of crack motion is very nearly perpendicular to the direction of the applied tensile stress and yields a relatively flat fracture surface as shown in Fig. 24(c).

- For most brittle crystalline materials, crack propagation corresponds to the successive and repeated breaking of atomic bonds along specific crystallographic planes (Figure 26); such a process is termed *cleavage*. This type of fracture is said to be **transgranular** (or *transcrystalline*) because the fracture cracks pass through the grains.

- In some alloys, crack propagation is along grain boundaries (Figure 27a); this fracture is termed **intergranular.** Figure 27*b* is a scanning electron micrograph showing a typical intergranular fracture, in which the three-dimensional nature of the grains may be seen. This type of fracture normally results subsequent to the occurrence of processes that weaken or embrittle grain boundary regions.

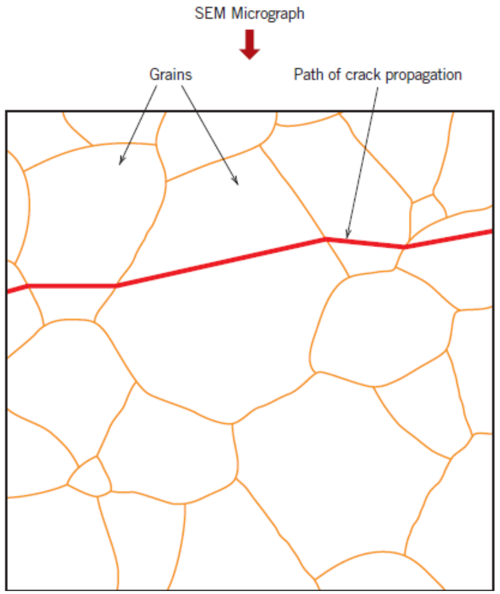
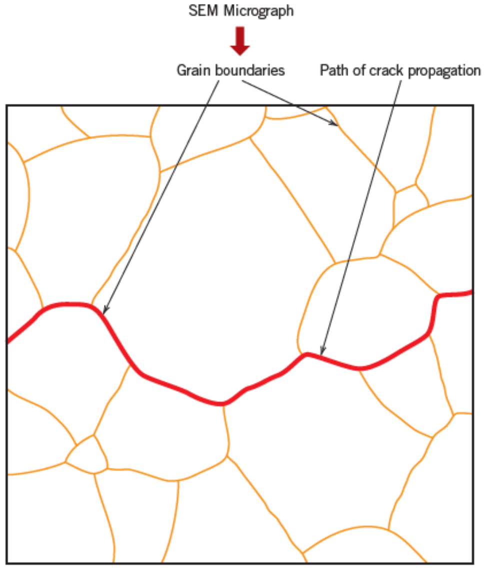
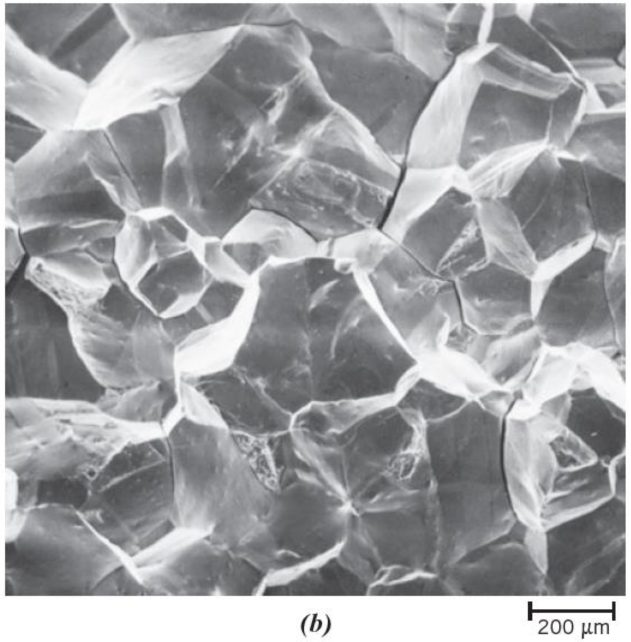


Figure 26 Schematic cross-section profile showing crack propagation through the interior of grains for transgranular fracture.





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PRINCIPLES OF FRACTURE MECHANICS

- **Fracture mechanics** allows quantifying the relationships among material properties, stress level, crack-producing flaws, and crack propagation mechanisms.

Stress Concentration:

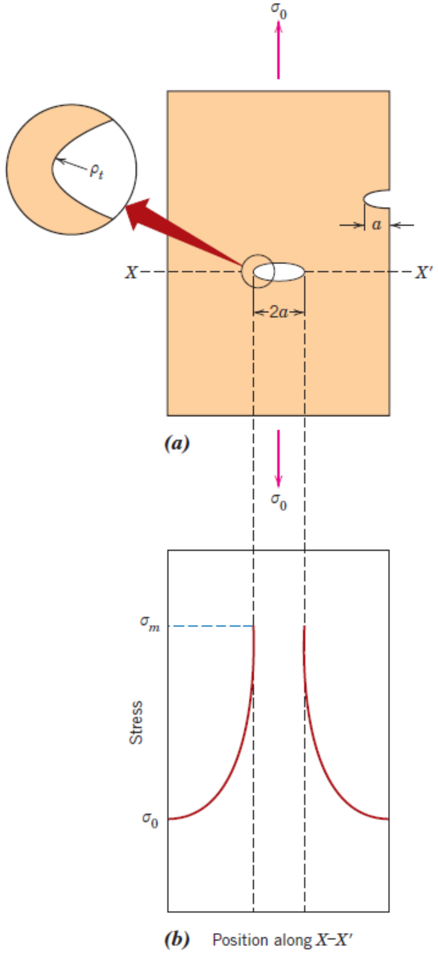
- The measured fracture strengths for most **brittle materials** are significantly **lower than** those predicted by **theoretical calculations** based on **atomic bonding energies**.
- This discrepancy is explained by the presence of **microscopic flaws or cracks** that always exist under normal conditions **at the surface** and **within the interior** of a body of material.
- These flaws are a detriment to the fracture strength because applied stress may be amplified or **concentrated at the tip**, the magnitude of this amplification depends on the **crack orientation and geometry**.

- A stress profile across a cross-section containing an internal crack. As indicated by this profile, the **magnitude of this localized stress decreases with distance away from the crack tip**.
- **At positions far removed**, the **stress** is just the **nominal stress σ_0** or the applied load divided by the specimen cross-sectional area (perpendicular to this load). Because of their ability to amplify applied stress in their locale, these flaws are sometimes called **stress raisers.**
- If it is assumed that a **crack** is similar to an **elliptical hole** through a plate and is oriented **perpendicular to the applied stress**, the maximum stress, σ_m, occurs at the **crack tip** and may be approximated by

$$\sigma_m = 2\sigma_0 \left(\frac{a}{\rho_t}\right)^{1/2}$$

where σ_0 is the magnitude of the nominal applied tensile stress, ρ_t is the radius of curvature of the crack tip (Figure 28*a*), and *a* represents the length of a surface crack or half of the length of an internal crack.

- For a relatively long microcrack that has a small tip radius of curvature, the factor $(a/\rho_t)^{1/2}$ may be very large. This yields a value of σ_m that is many times the value of σ_0. Sometimes the ratio σ_m/σ_0 is denoted by the ***stress concentration factor K_t***:



$$K_t = \frac{\sigma_m}{\sigma_0} = 2\left(\frac{a}{\rho_t}\right)^{1/2}$$

Figure 28 (*a*) The geometry of surface and internal cracks. (*b*) Schematic stress profile along the line *X*–*X′* in (*a*), demonstrating stress amplification at crack tip positions.

- **Stress amplification** is not **restricted** to these **microscopic defects**; it may occur at macroscopic **internal discontinuities** (e.g., voids or inclusions), **sharp corners**, **scratches,** and **notches**.

- Furthermore, the **effect of a stress raiser** is more **significant** in **brittle** than in **ductile materials**. For a ductile metal, **plastic deformation ensues** when the maximum stress **exceeds the yield strength**.

- This leads to a more uniform distribution of stress in the vicinity of the **stress raiser** and to the development of a **maximum stress concentration factor less than the theoretical value**.

- Such **yielding and stress redistribution** **does not occur** to any appreciable extent around flaws and discontinuities in **brittle materials**; therefore, essentially the **theoretical stress concentration results**.

- Using principles of fracture mechanics, it is possible to show that the critical stress $\boldsymbol{\sigma_c}$ required for crack propagation in a **brittle material** is described by the expression

$$\sigma_c = \left(\frac{2E\gamma_s}{\pi a}\right)^{1/2}$$

where E is the modulus of elasticity, γ_s is the specific surface energy, and a is one-half the length of an internal crack.

Impact Testing Techniques

- Impact testing techniques were established to ascertain the fracture characteristics of materials at high loading rates.
- For example, under some circumstances normally ductile metals fracture abruptly and with very little plastic deformation under high loading rates.
- Impact test conditions were chosen to represent those most severe relative to the potential for fracture—namely:

    i. deformation at a relatively low temperature,

    ii. a high strain rate (i.e., rate of deformation), and

    iii. a triaxial stress state (which may be introduced by the presence of a notch).

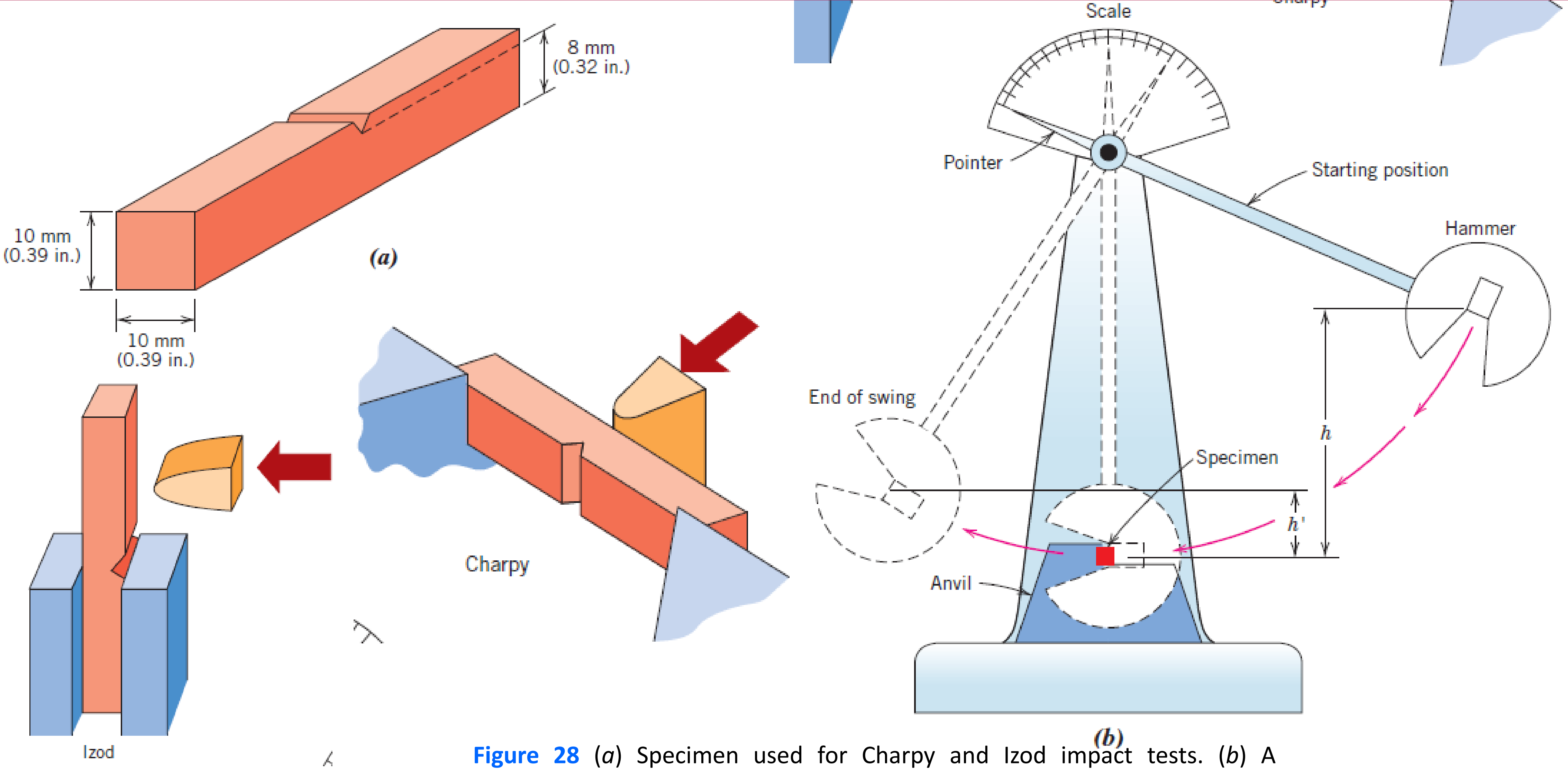


Figure 28 (*a*) Specimen used for Charpy and Izod impact tests. (*b*) A schematic drawing of an impact testing apparatus. The hammer is released from fixed height *h* and strikes the specimen; the energy expended in fracture is reflected in the difference between *h* and the swing height *h*′. Specimen placements for both the Charpy and the Izod tests are also shown.

Ductile-to-Brittle Transition

- One of the primary functions of the Charpy and the Izod tests is to determine whether a material experiences a **ductile-to-brittle transition** with decreasing temperature and, if so, the range of temperatures over which it occurs.

- The ductile-to-brittle transition is related to the temperature dependence of the measured impact energy absorption. This transition is represented for a steel by curve *A* in Figure 29.

- At higher temperatures, the CVN energy is relatively large, corresponding to a ductile mode of fracture.

- As the temperature is lowered, the impact energy drops suddenly over a relatively narrow temperature range, below which the energy has a constant but small value—that is, the mode of fracture is brittle.

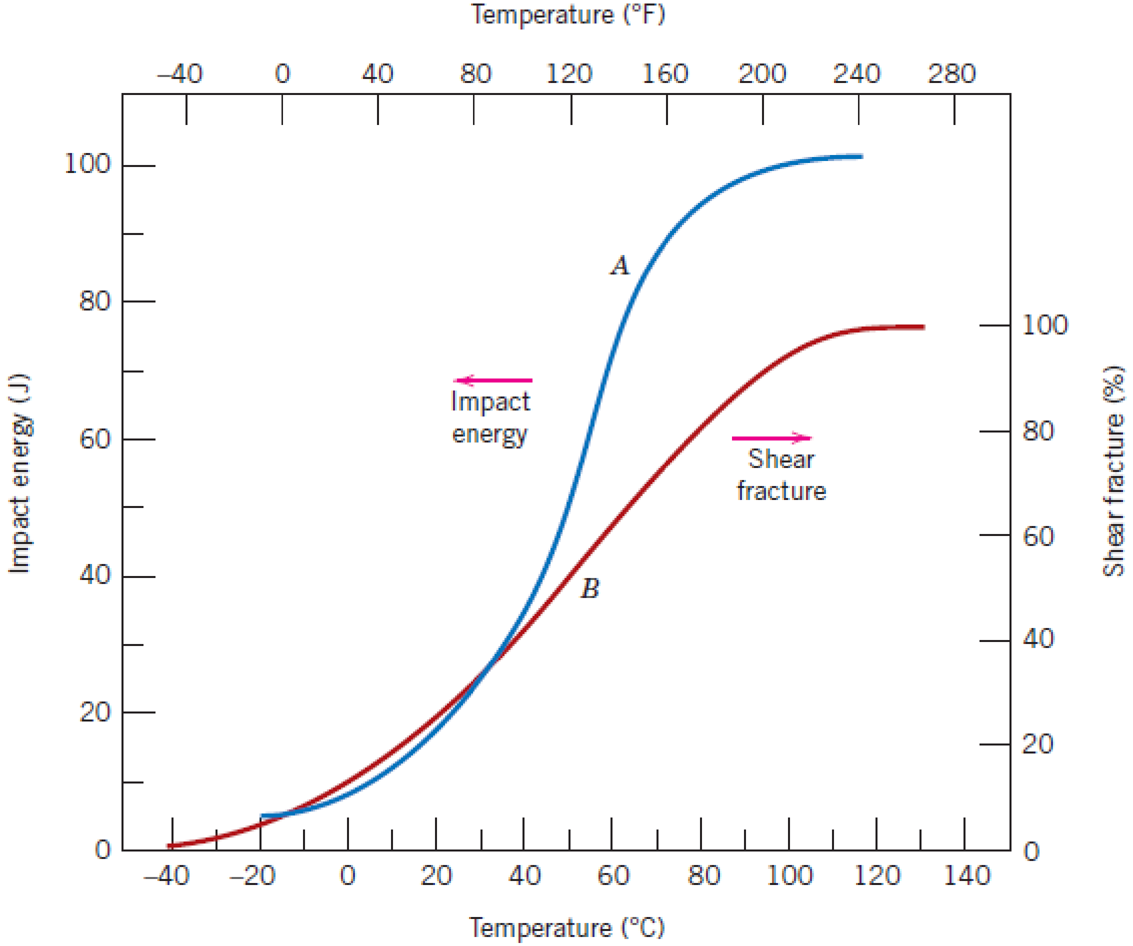


Figure 29 Temperature dependence of the Charpy V-notch impact energy (curve *A*) and percent shear fracture (curve *B*) for an A283 steel.

Fatigue:

- **Fatigue** is a form of failure that occurs in structures subjected to dynamic and fluctuating stresses (e.g., bridges, aircraft, and machine components).

- Under these circumstances, it is possible for failure to occur at a stress level considerably lower than the tensile or yield strength for a static load.

- The term *fatigue* is used because this type of failure normally occurs after a lengthy period of repeated stress or strain cycling.

- Fatigue failure is brittle-like in nature even in normally ductile metals, in that there is very little, if any, gross plastic deformation associated with failure.

- The process occurs by the initiation and propagation of cracks, and typically the fracture surface is perpendicular to the direction of an applied tensile stress.

CYCLIC STRESSES

- The applied stress may be axial (tension-compression), flexural (bending), or torsional (twisting) in nature. In general, three different fluctuating stress–time modes are possible.
- One is represented schematically by a regular and sinusoidal time dependence in Figure 30 *a*, where the amplitude is symmetrical about a mean zero stress level, for example, alternating from a maximum tensile stress (σ_{max}) to a minimum compressive stress (σ_{min}) of equal magnitude; this is referred to as a *reversed stress cycle*.
- Another type, termed a *repeated stress cycle*, is illustrated in Figure 30 *b*; the maxima and minima are asymmetrical relative to the zero stress level.
- Finally, the stress level may vary randomly in amplitude and frequency, as exemplified in Figure 30*c*.

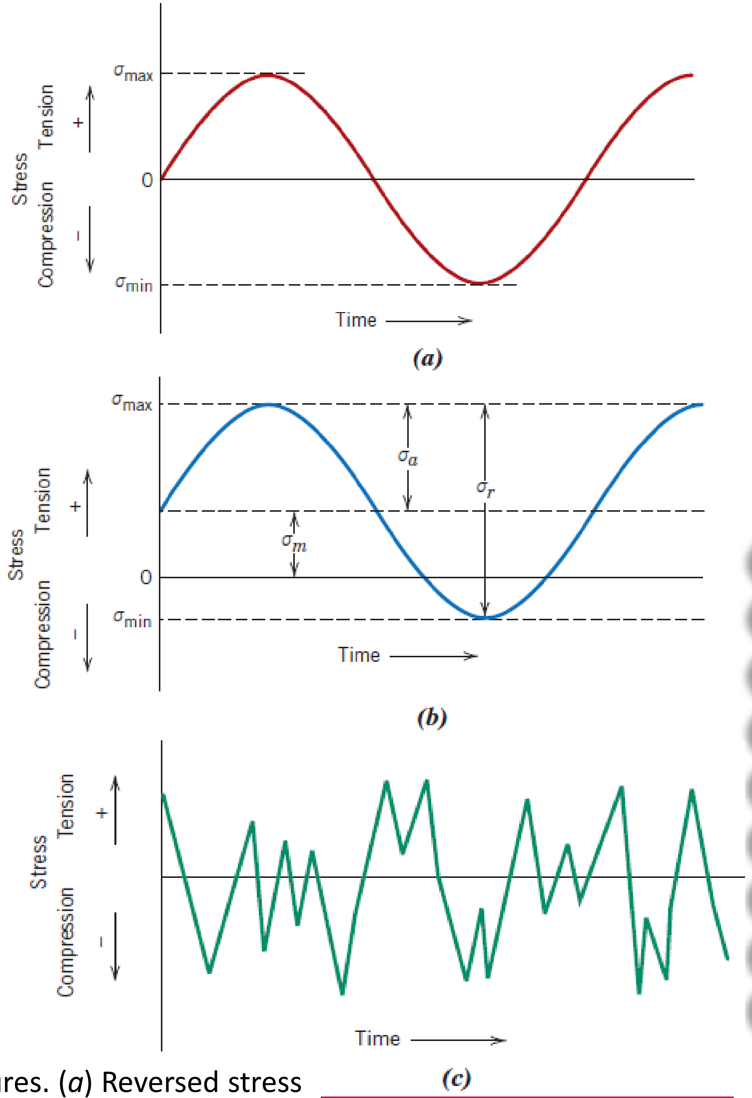


Figure 30 Variation of stress with time that accounts for fatigue failures. (*a*) Reversed stress cycle, in which the stress alternates from a maximum tensile stress (+) to a maximum compressive stress (–) of equal magnitude. (*b*) Repeated stress cycle, in which maximum and minimum stresses are asymmetrical relative to the zero-stress level; mean stress σm, range of stress σr, and stress amplitude σa are indicated. (*c*) Random stress cycle.

Mean stress for cyclic loading—dependence on maximum and minimum stress levels

$$\sigma_m = \frac{\sigma_{\max} + \sigma_{\min}}{2}$$

The *range of stress* σ_r is the difference between σ_{max} and σ_{min}, namely,

Computation of range of stress for cyclic loading

$$\sigma_r = \sigma_{\max} - \sigma_{\min}$$

Stress amplitude σ_a is one-half of this range of stress, or

Computation of stress amplitude for cyclic loading

$$\sigma_a = \frac{\sigma_r}{2} = \frac{\sigma_{\max} - \sigma_{\min}}{2}$$

Finally, the *stress ratio R* is the ratio of minimum and maximum stress amplitudes:

Computation of stress ratio

$$R = \frac{\sigma_{\min}}{\sigma_{\max}}$$

By convention, tensile stresses are positive and compressive stresses are negative. For example, for the reversed stress cycle, the value of *R* is –1.

THE *S–N* CURVE:

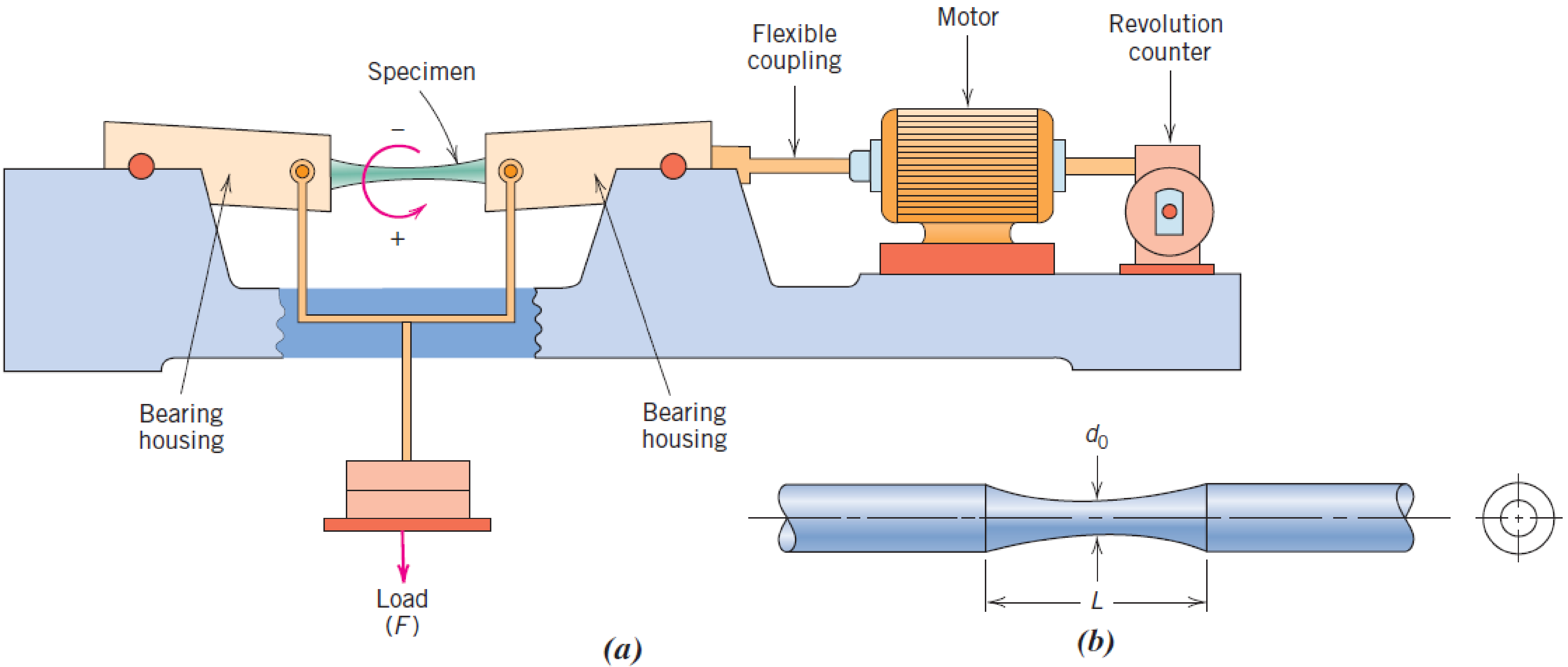


Figure 31 For rotating-bending fatigue tests, schematic diagrams of (*a*) a testing apparatus, and (*b*) a test specimen.

- The higher the magnitude of the stress amplitude, the smaller the number of cycles the material is capable of sustaining before failure.

- For some ferrous (iron base) and titanium alloys, the *S*–*N* curve (Figure 32) becomes horizontal at higher *N* values; there is a limiting stress level, called the **fatigue limit** (also sometimes called the *endurance limit*), below which fatigue failure will not occur.

- This fatigue limit represents the largest value of fluctuating stress that will *not* cause failure for essentially an infinite number of cycles.

- For many steels, fatigue limits range between 35% and 60% of the tensile strength.

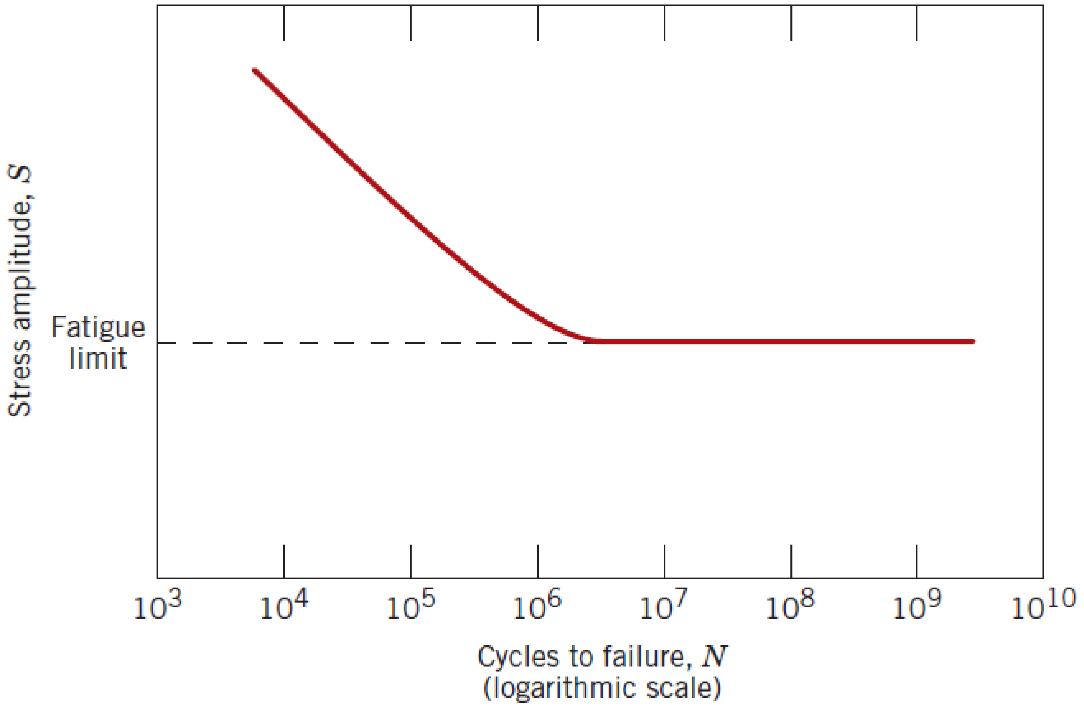


Figure 32 Stress amplitude (*S*) versus logarithm of the number of cycles to fatigue failure (*N*) for a material that displays a **fatigue limit**

- Most nonferrous alloys (e.g., aluminum, copper) do not have a fatigue limit, in that the *S–N* curve continues its downward trend at increasingly greater *N* values (Figure 33).

- Thus, fatigue ultimately occurs regardless of the magnitude of the stress. For these materials, one fatigue response is specified as **fatigue strength,** which is defined as the stress level at which failure will occur for some specified number of cycles (e.g., 10^7 cycles).

- The determination of fatigue strength is also demonstrated in Figure 33.

- Another important parameter that characterizes a material's fatigue behavior is **fatigue life** N_f.

- It is the number of cycles to cause failure at a specified stress level, as taken from the *S–N* plot (Figure 33).

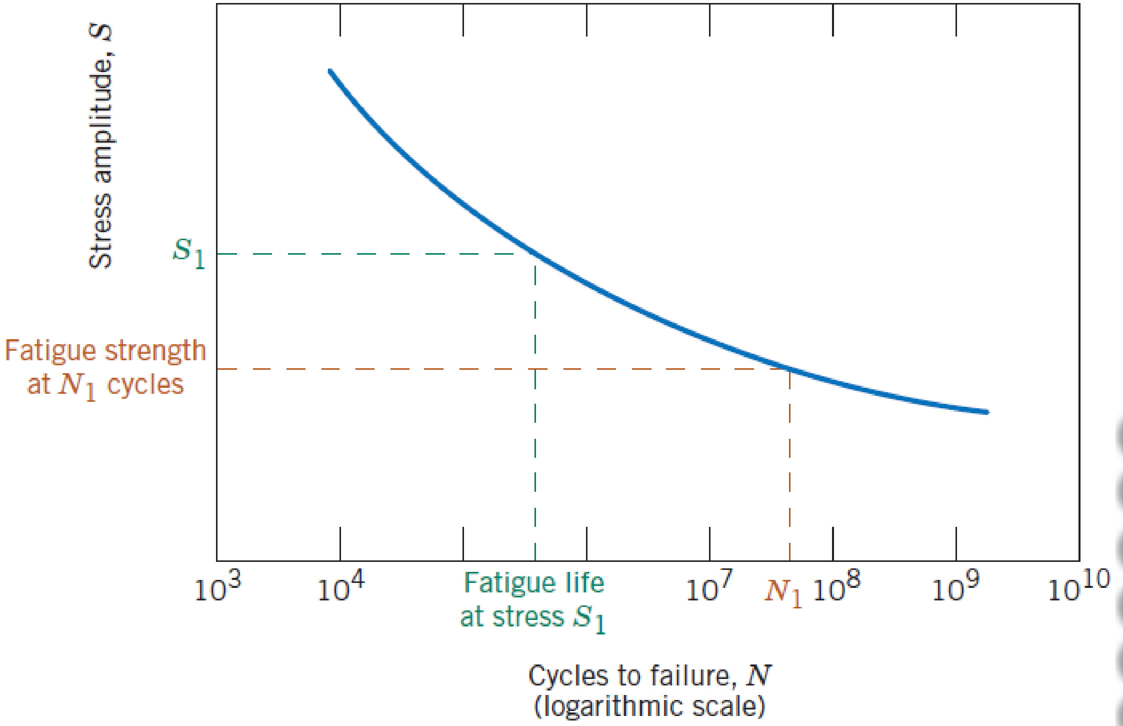


Figure 33 Stress amplitude (*S*) versus logarithm of the number of cycles to fatigue failure (*N*) for a material that **does not** display a **fatigue limit**

EXAMPLE PROBLEM

A cylindrical bar of **1045 steel** having the *S–N* behavior shown in Figure 34 is subjected to rotating–bending tests with reversed stress cycles. If the bar diameter is **15.0 mm**, determine the maximum cyclic load that may be applied to ensure that **fatigue failure will not occur**. Assume a **factor of safety of 2.0** and that the distance between load-bearing points is **60.0 mm**.

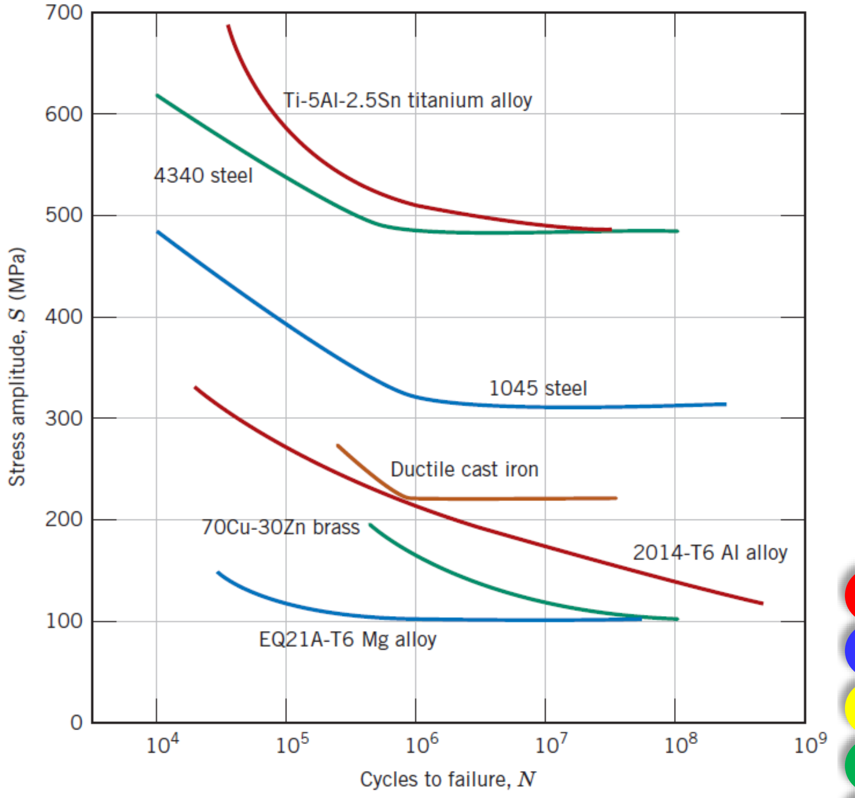


Figure 34 Stress amplitude (*S*) versus logarithm of the number of cycles to fatigue failure (*N*) for seven metal alloys.

Solution:

$$\sigma_a = \frac{\sigma_{max} - \sigma_{min}}{2} = \frac{\sigma_{max} + \sigma_{max}}{2} = \frac{2\sigma_{max}}{2} = \sigma_{max} \qquad \sigma_{max} = -\sigma_{min}$$

Hence, the maximum stress is also equal to the stress amplitude (310 MPa) from Fig. 34.

- Maximum stress for rotating-bending testes $\sigma_{max} = \dfrac{16LF_{max}}{\pi d_0^3}$

- F_{max} is the maximum applied load. When σ_{max} is divided by the factor of safety (N),

$$\frac{\sigma_{max}}{N} = \frac{16LF_{max}}{\pi d_0^3} \qquad F_{max} = \frac{\pi d_0^3 \sigma_{max}}{16NL}$$

$$F_{max} = \frac{(\pi)(15 \times 10^{-3}\ \mathrm{m})^3(310 \times 10^6\ \mathrm{N/m^2})}{(16)(2)(0.0600\ \mathrm{m})} = 1712\ \mathrm{N}$$

AMRITA VISHWA VIDYAPEETHAM School of Engineering
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EXAMPLE PROBLEM

A cylindrical **70Cu-30Zn brass bar** (Figure 34) is subjected to axial tension–compression stress testing with reversed cycling. If the load amplitude is **10,000 N**, **compute the minimum allowable bar diameter** to ensure that **fatigue failure will not occur at 10^7 cycles**. Assume a factor of safety of **2.5**.

Solution:

- From Figure 34, the fatigue strength (as stress amplitude) for this alloy at 10^7 cycles is 115 MPa (115×10^6 N/m^2);
- Tensile and compressive stresses are $$\sigma = \frac{F}{A_0}$$

Here, *F* is the applied load and A_0 is the cross-sectional area. For a cylindrical bar having a diameter of d_0,

$$A_0 = \pi\left(\frac{d_0}{2}\right)^2$$

$$\sigma = \frac{F}{A_0} = \frac{F}{\pi\left(\frac{d_0}{2}\right)^2} = \frac{4F}{\pi d_0^2}$$

replacing stress with the fatigue strength divided by the factor of safety

$$d_0 = \sqrt{\frac{4F}{\pi\left(\frac{\sigma}{N}\right)}} = \sqrt{\frac{(4)(10{,}000\ \text{N})}{(\pi)\left(\frac{115 \times 10^6\ \text{N/m}^2}{2.5}\right)}} = 16.6 \times 10^{-3}\ \text{m} = 16.6\ \text{mm}$$

Hence, the brass bar diameter must be at least 16.6 mm to ensure that fatigue failure will not occur.

Creep:

- Materials are often placed in service at elevated temperatures and exposed to static mechanical stresses (e.g., turbine rotors in jet engines and steam generators that experience centrifugal stresses; high-pressure steam lines).
- Deformation under such circumstances is termed **creep.**
- Defined as the time-dependent and permanent deformation of materials when subjected to a constant load or stress, creep is normally an undesirable phenomenon and is often the limiting factor in the lifetime of a part.
- It is observed in all materials types; for metals it becomes important only for temperatures greater than about 0.4*Tm*, where *Tm* is the absolute melting temperature.

GENERALIZED CREEP BEHAVIOR

- The resulting creep curve consists of three regions, each of which has its own distinctive strain–time feature. *Primary* or *transient creep* occurs first, typified by a continuously decreasing creep rate; that is, the slope of the curve decreases with time.

- For *secondary creep,* sometimes termed *steady-state creep*, the rate is constant—that is, the plot becomes linear. This is often the stage of creep that is of the longest duration. The constancy of creep rate is explained on the basis of a balance between the competing processes of strain hardening and recovery, recovery being the process by which a material becomes softer and retains its ability to experience deformation.

- Finally, for *tertiary creep*, there is an acceleration of the rate and ultimate failure.

- This failure is frequently termed *rupture* and results from microstructural and/or metallurgical changes—for example, grain boundary separation, and the formation of internal cracks, cavities, and voids.

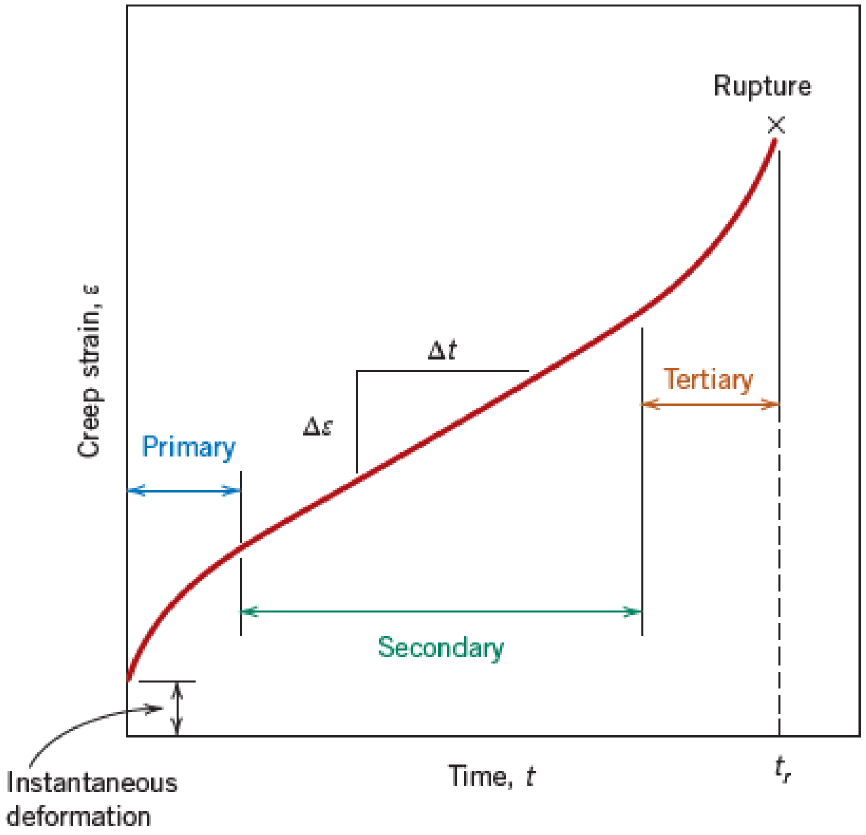

Figure 35 Typical creep curve of strain versus time at constant load and constant elevated temperature.

STRESS AND TEMPERATURE EFFECTS

- Both temperature and the level of the applied stress influence the creep characteristics (Figure 36).
- At a temperature substantially below 0.4*Tm*, and after the initial deformation, the strain is virtually independent of time.
- With either increasing stress or temperature, the following will be noted:
    i. The instantaneous strain at the time of stress application increases,
    ii. The steady-state creep rate increases, and
    iii. The rupture lifetime decreases.

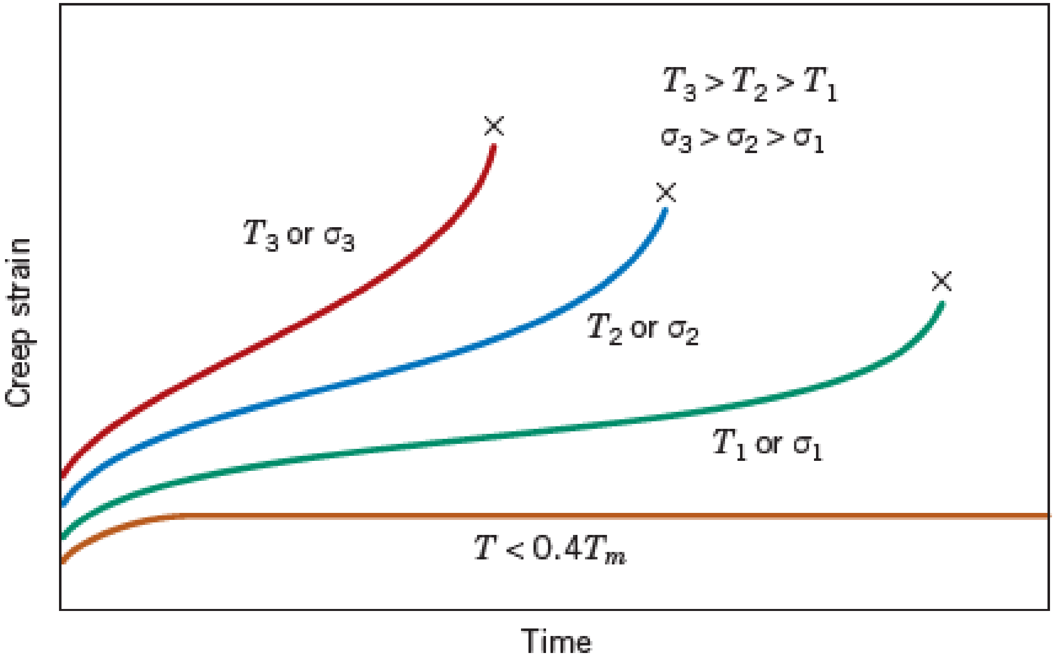

Figure 36 Influence of stress σ and temperature T on creep behavior.

$$\dot{\varepsilon}_s = K_1\sigma^n$$

AMRITA VISHWA VIDYAPEETHAM | School of Engineering Chennai Campus

Thank you !!!